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Deterioration under Abrupt Production Changes: Evaluating a DualLSTM with Bahdanau Attention against Statistical and Pretrained Benchmarks for Peruvian Agro-Industrial Forecasting under Small Data

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Abstract

Averaged forecast error can hide how models behave at abrupt production changes, which matters in short agricultural series. We evaluate a dual-branch LSTM with Bahdanau attention (DualLSTM), which processes the production history and a lagged exogenous context (climate, emergencies and news sentiment) separately, for one-step-ahead forecasts of national monthly production of sour lime and sweet lime in Peru. Under a leakage-controlled rolling-origin design with test years 2021–2025 (60 months per crop), it is compared with eight benchmarks, including the pretrained Chronos-Bolt-tiny, using the mean absolute error (MAE) over all months and over shock months (monthly P75 threshold) and the relative shock penalty Δ_s. For sour lime, Chronos had the lowest MAE (2,649 t; 95% CI 2,171–3,144), whereas the two DualLSTM variants had the highest (6,361 and 6,744 t). For sweet lime, SARIMA was the best model (18.4 t) and the only one ranked first or tied at all four thresholds. The DualLSTM showed low Δ_s (sour lime: -4.5% and 1.2%) only together with high absolute errors, and news features gave no clear benefit. Δ_s should therefore be reported together with absolute errors, and greater model complexity did not reduce the error with few data.

Keywords: agro-industrial forecasting; small data; DualLSTM; Bahdanau attention; abrupt production changes; shock-conditional evaluation; pretrained time-series models

1. Introduction

Forecasting monthly agricultural production is necessary to plan supply, logistics and prices, but in many countries the available series are short, strongly seasonal and subject to abrupt changes associated with climatic events, emergencies or market disruptions. With so little data, greater model complexity does not guarantee better out-of-sample performance: forecasting competitions and reviews of recurrent networks show that statistical and parsimonious methods remain hard benchmarks to beat [1–4], and it has been questioned whether deep architectures outperform simpler alternatives when data are scarce [5,6].

At the same time, the agricultural forecasting literature has proposed LSTM networks with attention mechanisms, hybrid models and boosting that incorporate meteorological and other exogenous variables [7–18]. Indicators derived from news and the integration of heterogeneous sources have also been used as auxiliary information [19–23]. However, many of these studies use longer series, compare against few simple benchmarks, or

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evaluate only the average error, without examining what happens in the months of abrupt change, which are those of greatest operational interest. In addition, pretrained time-series models such as Chronos [24], TimesFM [25] and Moirai [26] offer a new benchmark that can forecast without local training.

This work extends a preliminary study presented at IEEE LATINCOM 2026 [27]. It evaluates a DualLSTM architecture with Bahdanau attention that explicitly separates the production history from the lagged exogenous context. The aim is not to demonstrate the superiority of the architecture, but to measure, under a protocol that controls information leakage, how it compares with statistical, regularised, tree-based and pretrained benchmarks on two Peruvian national series: sour lime and sweet lime. The research questions are:

1. Does the DualLSTM with attention outperform simpler and pretrained benchmarks in overall forecast error with few data?
2. Do the conclusions hold when the evaluation is restricted to months of abrupt change?
3. Do variables derived from news text improve forecast accuracy?

The contributions are: (i) a reproducible rolling-origin evaluation of a dual-branch recurrent architecture against eight benchmarks on two crops; (ii) an analysis conditioned on abrupt changes, with a threshold defined only from information available at the forecast origin and its sensitivity; (iii) a multi-seed evaluation without selection of the best seed; and (iv) an explicit account of which parts of the protocol were fixed after seeing earlier results, so that the evidence is not over-read. [PENDIENTE: verify the TimesFM and Moirai references (PMLR 235 pages) in Scopus.]

2. Materials and Methods

2.1. Study design

We evaluate one-step-ahead forecasts of monthly national production of two crops, sour lime (*limón sutil*) and sweet lime (*limón dulce*), in Peru, using a rolling-origin design [28] (Figure 1). For each test year $Y \in \{2018, \ldots, 2025\}$ the forecast origin is December of $Y-1$: models are fitted with data up to December of $Y-2$, tuned and early-stopped on year $Y-1$, and produce twelve one-step-ahead forecasts for year Y (Figure 6). Lagged values of the test year are taken from observations, as is usual in one-step-ahead evaluation. All models share the same partitions in every window.

Trained models require at least 36 training months. Windows 2018–2020 therefore include only the three models that need no training (Naive, Seasonal Naive and Chronos), which gives 118 viable model–window–crop combinations and 42 that do not apply. The primary ranking pools the windows 2021–2025; a sensitivity analysis repeats all conclusions on 2023–2025.

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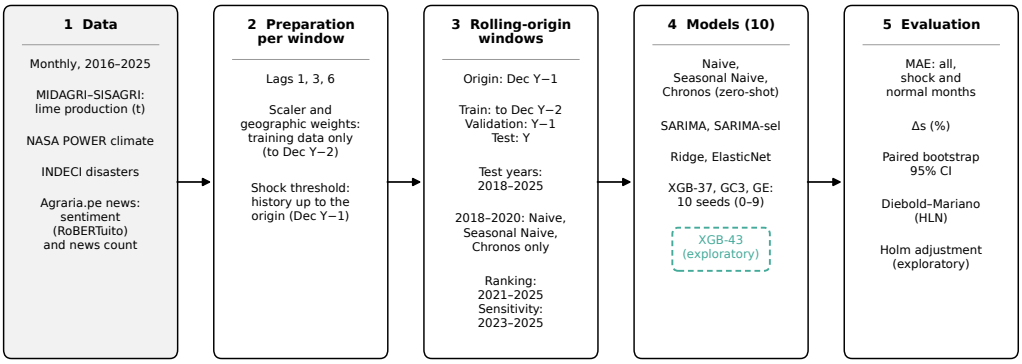


Figure 1. Study workflow: data sources, preparation within each window, rolling-origin windows, models and evaluation. The dashed box marks the exploratory XGB-43 model, which is outside the ranking.

2.2. Data

The national series of each crop is the sum of monthly provincial production recorded by MIDAGRI–SISAGRI [29] from January 2016 to December 2025 (109 provinces with a record of at least one of the two crops; Figure 2). Exogenous variables enter the models only with lags of 1, 3 and 6 months.

- **Climate:** NASA POWER monthly values at the coordinates of the 109 provinces: mean (T2M) and maximum (T2M_MAX) temperature, wind speed (WS2M), precipitation (PRECTOTCORR) and relative humidity (RH2M). Units: temperature in °C (T2M is the monthly mean and T2M_MAX the monthly mean of the daily maximum), wind speed in $\mathrm{m\,s^{-1}}$, precipitation in $\mathrm{mm\,day^{-1}}$ (monthly mean of daily totals, not a monthly sum) and relative humidity in %. No values were missing and no imputation was applied. Ten provinces use the reference coordinate of their department.
- **Emergencies:** the official INDECI emergency records, restricted to ten climate-related hazards (heavy rain, strong winds, low temperatures, flood, landslide, mudflow, drought, erosion, thunderstorm and swell). Five monthly series are used: number of emergencies, people affected, people homeless, and hectares of crop lost and affected. Months without records are coded as zero. The conversion of the raw INDECI records into this monthly panel, including the zero coding, has no associated code in the repository (see Reproducibility).
- **News sentiment:** 2,883 news items from the agribusiness section of Agraria.pe. The sentiment of each item is $s = P(\text{positive}) - P(\text{negative})$ from the RoBERTuito model of `pysentimiento` [30,31], applied zero-shot to the headline and the first 200 words. Two monthly series are built (mean sentiment and number of items); months without news (January and February 2016) are set to zero.

Provincial climate and emergency series are aggregated to national level with fixed, crop-aware geographic weights estimated only on the training period of each window. When a province has no records, the aggregator renormalises over the weight mass present. Lags are computed on the already aggregated national series.

Table 2 summarises the series of each window, with the ADF [32] and KPSS [33] unit-root and stationarity tests.

2.3. Shock definition

For each month the absolute month-on-month relative change is

$$r_t = \left| \frac{y_t - y_{t-1}}{y_{t-1}} \right|, \tag{1}$$

and the year-on-year version replaces y_{t-1} by y_{t-12}. A month of the test year is a *shock* if $r_t > P_k$, where P_k is the linear-interpolation percentile of the changes dated from the start of the series up to the forecast origin. The primary definition is the 75th percentile (P75) of the month-on-month change; P50, P60 and P90, and the year-on-year version, are reported as sensitivity points, and a full sweep from P0 to P95 is computed. P0 (every month is a shock) is used as a code check, because Δ_s must equal exactly zero there. A window with fewer than two shock months is flagged as non-informative, but its months are kept in the pooled estimates. The definition is statistical and attributes no cause.

2.4. Models

Ten models are ranked (Table 1).

Table 1. Models in the ranking. Seeded models use ten seeds (0–9).

Model	Description
Naive, Seasonal Naive	$\hat{y}_t = y_{t-1}$ and $\hat{y}_t = y_{t-12}$
SARIMA	Fixed order, one-step updating without re-estimation [34]
SARIMA-sel	Order selected in each window (see below)
Chronos	Pretrained Chronos-Bolt-tiny [24], zero-shot, univariate
Ridge, ElasticNet	Regularised linear models with 37 predictors [35]
XGB-37	Gradient boosting with the same 37 predictors [36]
GC3	Dual-LSTM with attention, 37 predictors (no news)
GE	GC3 plus six lagged news features

Benchmarks without training. Chronos uses `amazon/chronos-bolt-tiny` with the median forecast, horizon 1 and the full univariate history from January 2016 up to $t-1$. Its comparison with the multivariate models concerns predictive performance, not equal information sets.

SARIMA. The fixed orders are $(1,1,1)(1,1,0)_{12}$ for sour lime and $(1,0,0)(0,1,0)_{12}$ for sweet lime, both with a constant term, seasonal period 12 and one-step-ahead updating without re-estimating parameters. For SARIMA-sel, a grid of 576 orders ($p \in 0..3$, $d \in 0..1$, $q \in 0..3$, $P \in 0..2$, $D \in 0..1$, $Q \in 0..2$, $m = 12$) is fitted on the training months; the five orders with the lowest AIC are refitted, and the one with the lowest one-step-ahead MAE on year $Y-1$ is kept.

Ridge and ElasticNet. The penalty $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ and, for ElasticNet, the mixing ratio $\{0.2, 0.5, 0.8\}$ are re-chosen in each window by the validation MAE in tonnes; ties are resolved in favour of the first grid entry. The final fit uses the training months only. Predictors are standardised with the training mean and standard deviation.

XGB-37. Squared-error gradient boosting with maximum depth 2, learning rate 0.05, 200 trees, subsample 0.8 and column subsample 0.8, with default regularisation and no hyperparameter search. The 37 predictors are the four production variables (current production and its lags 1, 3 and 6), three calendar terms (sine and cosine of the month and a time index), and the five climate and five emergency series at lags 1, 3 and 6.

Dual-LSTM with attention (GC3 and GE). Two recurrent branches process the previous six months (Figure 7). Branch A receives the four production variables and branch B the calendar, climate and emergency variables (33 features in GC3; 39 in GE, which adds mean sentiment and news count at lags 1, 3 and 6). Each branch has a 16-unit LSTM [37] followed by additive Bahdanau attention [38] of 16 units; the context vectors are concatenated and passed through dense layers of 16 and 8 units (ReLU), dropout 0.20 and a scalar output. GC3 has 6,273 and GE 6,657 trainable parameters. Training uses Adam [39] (learning rate 0.001), mean-squared-error loss, batch size 8, up to 300 epochs, no shuffling, early stopping

on the validation year (patience 15, best weights restored) and learning-rate reduction on plateau (factor 0.5, patience 8). No hyperparameter search was run.

Seeds. XGB-37, GC3 and GE are trained with ten seeds (0–9) and no seed is selected. The error series of a seeded model is the monthly mean of the absolute errors over its seeds, and seed dispersion is reported separately. Each network run starts in a new process with deterministic settings.

Exploratory model. XGB-43 adds the six lagged news features to XGB-37 with the same hyperparameters. It was added post hoc, is not part of the ranking, the Diebold–Mariano tests or the multiplicity correction, and appears only in the exploratory comparison with XGB-37.

2.5. Leakage control

Within each window, the scaler, the geographic weights, the hyperparameters that are re-chosen (Ridge, ElasticNet, SARIMA-sel) and the MASE denominator use data up to December of $Y-2$ only, and the shock threshold uses the history up to the forecast origin (December of $Y-1$). Early stopping and model selection use year $Y-1$, never the test year. An automated test checks these conditions.

2.6. Evaluation

The main metric is the mean absolute error (MAE, in tonnes) over all test months, over the shock months and over the remaining (normal) months. The relative shock penalty is

$$\Delta_s = \frac{\mathrm{MAE}_{\mathrm{shock}} - \mathrm{MAE}_{\mathrm{all}}}{\mathrm{MAE}_{\mathrm{all}}} \times 100. \tag{2}$$

Because Δ_s is a ratio, it can be low either because the shock error is small or because the normal-month error is large; it is therefore always reported together with $\mathrm{MAE}_{\mathrm{shock}}$ and $\mathrm{MAE}_{\mathrm{normal}}$. We also compute the RMSE, the ratio of shock to normal MAE, Δ_s when each shock month is removed in turn, the MAE by quartile of $|\Delta y/y|$, and the MASE with a single denominator per window [40], the mean of $|y_t - y_{t-12}|$ over the training months of that window (undefined in 2018). A model is called best if it ranks first, or ties with the first, in at least three of the four reporting thresholds P50, P60, P75 and P90 for a crop.

Uncertainty. Confidence intervals come from a paired bootstrap [41] over months with the same indices for all models, stratified by window ($B = 10{,}000$, seed 42, 2.5th and 97.5th percentiles). For $\mathrm{MAE}_{\mathrm{shock}}$ and $\mathrm{MAE}_{\mathrm{normal}}$, months are resampled within their own type. A sensitivity analysis uses circular moving blocks of three months. A model ties with the first if the interval of its difference with the first contains zero.

Significance tests. Pairs of models are compared with the Diebold–Mariano test with the Harvey–Leybourne–Newbold correction [42,43] ($h = 1$, absolute loss, seed-averaged errors, pooled windows 2021–2025, two-sided t_{n-1}), on all months and on shock months only. With few shock months, power is low. As an exploratory addition, p-values are adjusted by the Holm method [44] within each family (crop $\times$ subset, 45 pairs).

Fit diagnostics. A model–window is flagged as overfitted if the ratio of validation to training MAE exceeds 2.0; the same ratio for the Naive forecast is shown for reference.

2.7. Transparency about the protocol

The protocol was written down before the rolling-origin experiment was run, but the test year 2025 had already been inspected in earlier experiments of this project. The hyperparameters of GC3, GE and XGB-37 were fixed with data up to 2024, and the order of the fixed sour-lime SARIMA was adopted after observing the 2025 results. Windows

2021–2024 are therefore retrospective evaluations, not independent test sets. We report these facts so that the 2025 window is not read as a confirmatory test.

2.8. Reproducibility

All experiments ran on CPU only (Windows 11 Pro; AMD64 family 25, model 33 processor with 16 logical cores) with Python 3.11.9, TensorFlow 2.21.0, Keras 3.14.0, statsmodels 0.14.6, xgboost 3.2.0, scikit-learn 1.8.0, torch 2.11.0 (CPU) and chronos-forecasting 2.3.2, with `pysentimiento` 0.7.3. Environment variables fix the hash seed and deterministic operations, the global seed is 42, and the SHA-256 hashes of the frozen input tables are stored with the code. Everything from the frozen tables to the final metrics is reproducible with a single driver script. Three links of the data chain have no code in the repository and are provided as frozen tables: the processing of the MIDAGRI–SISAGRI spreadsheet into national series, the processing of the INDECI records into the monthly panel, and the province coordinates. The news corpus and the NASA POWER values can change if collected again. Code and frozen tables: `https://github.com/YeralD21/machine-learning-multimodal-agro-nlp-clima-` [PENDIENTE: create a release tag after pushing the final code]. The MIDAGRI–SISAGRI table was downloaded from the Regional Productive Profile dashboard on 25 June 2026 (the file covers February 2015 to March 2026; this study uses January 2016 to December 2025); because the dashboard is updated monthly and past figures may be revised, the frozen copy used here, with its SHA-256 hash, is provided in the repository.

3. Results

3.1. Series and overall accuracy

Table 2 describes the series by window. Sour lime production averages between 23,067 and 32,209 t per month in the test years, with a coefficient of variation of 0.16–0.36, and sweet lime between 394 and 468 t, with 0.36–0.39. The unit-root tests are not conclusive: for sour lime the ADF test rejects a unit root in 2021–2023 but not in 2024–2025, where the KPSS test rejects level stationarity, and for sweet lime neither test rejects its null in any window. Figure **??** shows the series and the shock months.

Table 3 gives the pooled accuracy over the 60 test months of 2021–2025. For sour lime, Chronos had the lowest MAE (2,649 t; 95% CI 2,171–3,144), followed by SARIMA-sel (3,206 t), SARIMA (3,361 t) and the Naive forecast (3,624 t). GC3 and GE had the highest errors of the ten models (6,361 and 6,744 t), above the Seasonal Naive (6,075 t). For sweet lime, SARIMA had the lowest MAE (18.4 t; 14.5–22.8), followed by SARIMA-sel (20.6 t) and the Seasonal Naive (24.5 t), whereas GC3 and GE both reached 56.7 t, above Ridge, ElasticNet and XGB-37 (48.3–53.1 t). Figure 25 shows the 2025 forecasts.

3.2. Shock months

With the monthly P75 threshold, the pooled sample contains 10 shock months for sour lime and 15 for sweet lime (Figure 22). In sour lime, Chronos also had the lowest MAE in shock months (4,784 t), and SARIMA-sel and the Seasonal Naive ranked second and third (Table 4); GC3 ranked sixth and GE seventh. In sweet lime, SARIMA ranked first (14.0 t), with the Seasonal Naive third, GE fifth and GC3 sixth. The intervals for shock months are wide (for example 2,259–7,928 t for the Seasonal Naive in sour lime), so many models are statistically tied with the first (asterisks in Table 4).

The penalty Δ_s must be read together with the absolute errors (Table 3, Figure 10). In sour lime, the Seasonal Naive had $\Delta_s = -17.1\%$ and GC3 -4.5%, but both had a high MAE in all months, so their shock error was not low in absolute terms: Chronos, with $\Delta_s = 80.6\%$, had a lower shock MAE than either. In sweet lime, SARIMA combined the

lowest shock error with $\Delta_s = -24.1\%$, GC3 and GE had Δ_s of 7.2% and 2.0%, and the Seasonal Naive had $\Delta_s = 0.0\%$.

A model is robust if it is first or tied in at least three of the four thresholds P50, P60, P75 and P90 (Table 4; Figures 19, 16 and 13). With the month-on-month definition, Chronos, SARIMA and SARIMA-sel were robust in sour lime, and only SARIMA in sweet lime. Neither GC3 nor GE was robust in either crop.

3.3. Statistical comparisons

Tables 11 and 12 report the Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction (unadjusted p-values; Holm-adjusted values in Figure 40, exploratory). On all months of sour lime, GC3 and GE had significantly larger errors than Chronos, SARIMA, SARIMA-sel and XGB-37 ($p \leq 0.001$). With only 10 shock months, the only comparisons with $p < 0.05$ involved the Naive forecast. In sweet lime, SARIMA had a significantly lower error than every model except SARIMA-sel on all months ($p = 0.20$ against SARIMA-sel), and on the 15 shock months it was significantly better than every other model ($p \leq 0.001$) except SARIMA-sel ($p = 0.115$). Power is limited in all shock-only comparisons.

3.4. Contribution of news features

Table 13 and Table 14 (Figure 28) compare GE with GC3, seed by seed. In sour lime, adding the news features increased the pooled error by 382 t (95% CI 45–711), and GE was better than GC3 in only one of ten seeds. In sweet lime the difference was 0.1 t (95% CI −1.9 to 1.9), with no evidence of an effect. The exploratory comparison of XGB-43 with XGB-37 (Table 6; post hoc, outside the ranking) found a higher error in sour lime (130.5 t; CI 1.5–260.4) and, in sweet lime, no difference over all months (−0.6 t; CI −2.4 to 1.2) but a lower shock error (−3.6 t; CI −5.9 to −1.2, 15 months).

3.5. Training diagnostics

Loss curves, early-stopping epochs and the training–validation gap of the networks are shown in Figures 46, 49, 52 and 31; the dispersion across seeds is in Figure 73. Per-window MAE values are in Tables 7 to 10.

Table 2. Descriptive statistics of the national monthly production series for each rolling-origin window, 2021–2025. Mean, standard deviation (SD) and coefficient of variation (CV) refer to the 12 months of the test year. Train months, autocorrelation (ACF at lags 1 and 12) and unit-root tests refer to the training partition of each window, series in levels with a constant: ADF [32] (null hypothesis: unit root) and KPSS [33] (null hypothesis: level stationarity) p-values; KPSS p-values outside the tabulated range are shown as bounds.

Test year	Train months	Mean (t)	SD (t)	CV	ACF(1)	ACF(12)	ADF p	KPSS p
Sour lime								
2021	42	27,980	4,362	0.16	0.84	0.33	0.011	>0.10
2022	54	30,558	8,004	0.26	0.81	0.49	0.004	>0.10
2023	66	23,067	8,319	0.36	0.84	0.55	0.002	0.035
2024	78	32,209	8,550	0.27	0.86	0.56	0.863	0.016
2025	90	31,908	9,103	0.29	0.85	0.58	0.641	0.039
Sweet lime								
2021	42	393.8	144.0	0.37	0.79	0.67	0.267	>0.10
2022	54	406.4	154.5	0.38	0.80	0.74	0.915	>0.10
2023	66	406.5	159.9	0.39	0.81	0.79	0.779	>0.10
2024	78	448.3	161.4	0.36	0.81	0.81	0.446	>0.10
2025	90	467.8	177.3	0.38	0.82	0.83	0.377	>0.10

Table 3. Aggregated forecast accuracy, pooled test windows 2021–2025, by cultivar. Shock months: monthly P75 threshold of the month-on-month change (history up to each origin). MAE in tonnes with 95% bootstrap confidence intervals [41] in parentheses (months resampled within windows, B = 10,000); $\Delta_s = (\mathrm{MAE}_{\mathrm{shock}} - \mathrm{MAE})/\mathrm{MAE} \times 100$. Errors of seeded models (XGB-37, GC3, GE) are averaged over 10 seeds per month. The exploratory XGB-43 is not included.

Model	MAE, all months (t) (95% CI)	MAE, shock months (t) (95% CI)	Δ_s (%)
Sour lime (60 test months; 10 shock months)			
Naive	3,624 (2,920–4,382)	8,506 (7,724–9,215)	134.7
Seasonal Naive	6,075 (5,156–6,987)	5,034 (2,259–7,928)	-17.1
Chronos	2,649 (2,171–3,144)	4,784 (3,963–5,641)	80.6
SARIMA	3,361 (2,780–3,970)	5,131 (3,759–6,624)	52.7
SARIMA-sel	3,206 (2,563–3,880)	4,936 (2,866–7,242)	54.0
Ridge	4,925 (3,838–6,074)	8,557 (5,484–12,360)	73.7
ElasticNet	4,872 (3,847–5,950)	8,154 (5,162–11,484)	67.4
XGB-37	4,881 (4,063–5,732)	5,955 (3,727–7,965)	22.0
GC3	6,361 (5,292–7,446)	6,074 (4,085–8,216)	-4.5
GE	6,744 (5,587–7,894)	6,827 (4,477–9,213)	1.2
Sweet lime (60 test months; 15 shock months)			
Naive	77.5 (65.7–89.9)	131.6 (116.5–147.7)	69.8
Seasonal Naive	24.5 (20.5–29.0)	24.5 (21.4–27.8)	-0.0
Chronos	30.8 (25.6–36.4)	48.2 (40.5–56.1)	56.4
SARIMA	18.4 (14.5–22.8)	14.0 (9.8–18.0)	-24.1
SARIMA-sel	20.6 (16.5–25.2)	19.7 (15.6–23.6)	-4.5
Ridge	53.1 (43.4–63.4)	69.7 (52.0–89.3)	31.3
ElasticNet	50.2 (40.7–60.4)	67.2 (49.5–87.1)	33.8
XGB-37	48.3 (39.8–57.2)	65.8 (51.6–80.6)	36.4
GC3	56.7 (46.9–66.9)	60.7 (46.2–76.3)	7.2
GE	56.7 (47.5–66.7)	57.9 (41.8–75.2)	2.0

Table 4. Rank of each model by MAE in shock months, pooled test windows 2021–2025, at the reporting thresholds P50, P60, P75 and P90 of the month-on-month (MoM) change. * = first or statistically tied with the first (the 95% paired bootstrap CI of the difference in MAE with the best model contains 0). A model is robust if it is first or tied in at least 3 of the 4 thresholds; the year-on-year (YoY) definition is evaluated separately and its ranks are not shown.

Model	P50	P60	P75	P90	First or tied, MoM (of 4)	Robust, MoM	First or tied, YoY (
Sour lime							
Naive	5	8	9	7	0	No	
Seasonal Naive	6	5	3*	4*	2	No	
Chronos	2*	3*	1*	1*	4	Yes	
SARIMA	3*	2*	4*	2	3	Yes	
SARIMA-sel	1*	1*	2*	3	3	Yes	
Ridge	9	10	10	10	0	No	
ElasticNet	10	9	8	9	0	No	
XGB-37	4	4	5*	6	1	No	
GC3	8	6	6*	5	1	No	
GE	7	7	7	8	0	No	
Sweet lime							
Naive	10	10	10	10	0	No	
Seasonal Naive	3	3	3	2	0	No	
Chronos	4	4	4	4	0	No	
SARIMA	1*	1*	1*	1*	4	Yes	
SARIMA-sel	2*	2*	2	3	2	No	
Ridge	9	9	9	8	0	No	
ElasticNet	8	8	8	7	0	No	
XGB-37	5	6	7	9	0	No	
GC3	7	7	6	5	0	No	
GE	6	5	5	6	0	No	

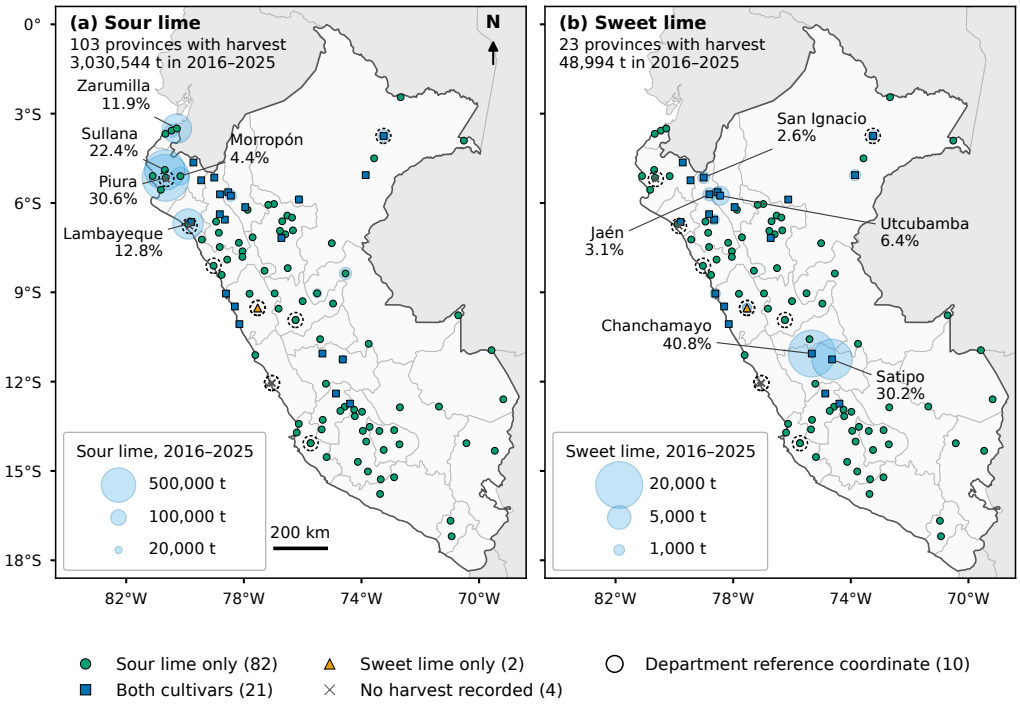


Figure 2. Lime-producing provinces of Peru and their cumulative production, January 2016–December 2025: (a) sour lime, (b) sweet lime (MIDAGRI–SISAGRI (accessed [access date])). Circle area is proportional to each province's cumulative production of that cultivar; labels give the five largest provinces and their share of the national total of that cultivar. Symbols show whether a province recorded sour lime only, both cultivars, sweet lime only or no harvest in the period (109 provinces). Dashed rings: ten provinces placed at their department's reference coordinate, including the two sweet-lime-only and the four no-harvest provinces. Boundaries: geoBoundaries gbOpen [45], Public Domain, 2008 vintage; neighbouring countries: Natural Earth 1:10m.

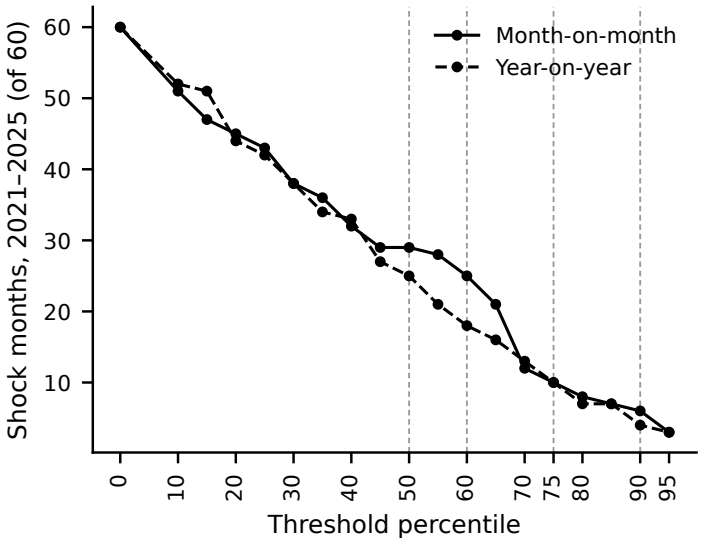


Figure 3. Sour lime

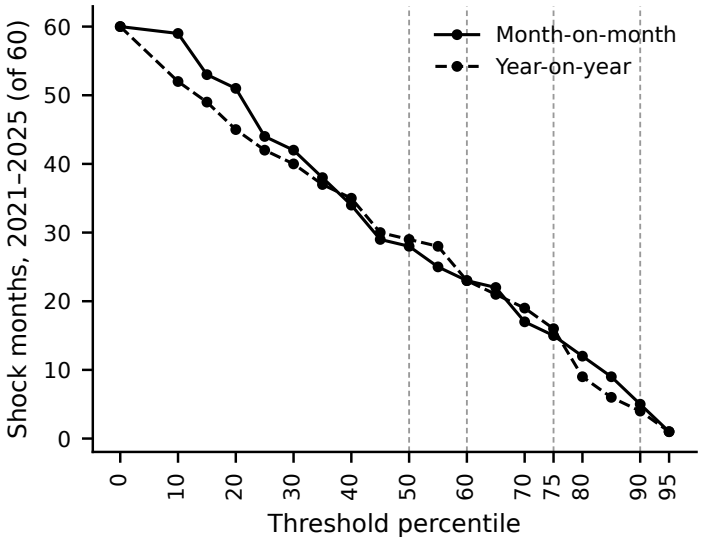


Figure 4. Sweet lime

Figure 5. *Sour lime:* Number of shock months in the pooled 2021–2025 test set versus threshold percentile, sour lime (limón sutil); solid: month-on-month change, dashed: year-on-year change. *Sweet lime:* Number of shock months in the pooled 2021–2025 test set versus threshold percentile, sweet lime (limón dulce); solid: month-on-month change, dashed: year-on-year change.

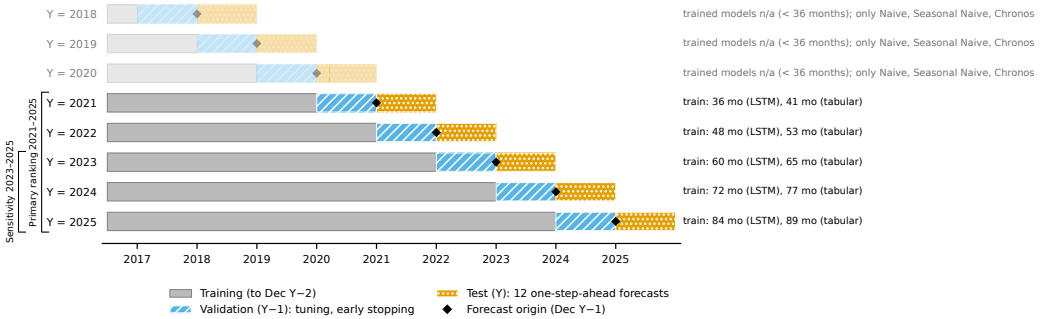


Figure 6. Rolling-origin evaluation design [28]. For each test year Y (2018–2025) the forecast origin is December of Y−1; trained models use data up to December of Y−2 (features from 2016-07; tabular targets from 2016-08; LSTM targets from 2017-01), are tuned and early-stopped on year Y−1, and produce twelve one-step-ahead forecasts for year Y. Trained models require at least 36 training months, so windows 2018–2020 (faded) include only the untrained models (Naive, Seasonal Naive, Chronos). The primary ranking pools the windows 2021–2025; a pre-registered sensitivity analysis repeats all conclusions on 2023–2025.

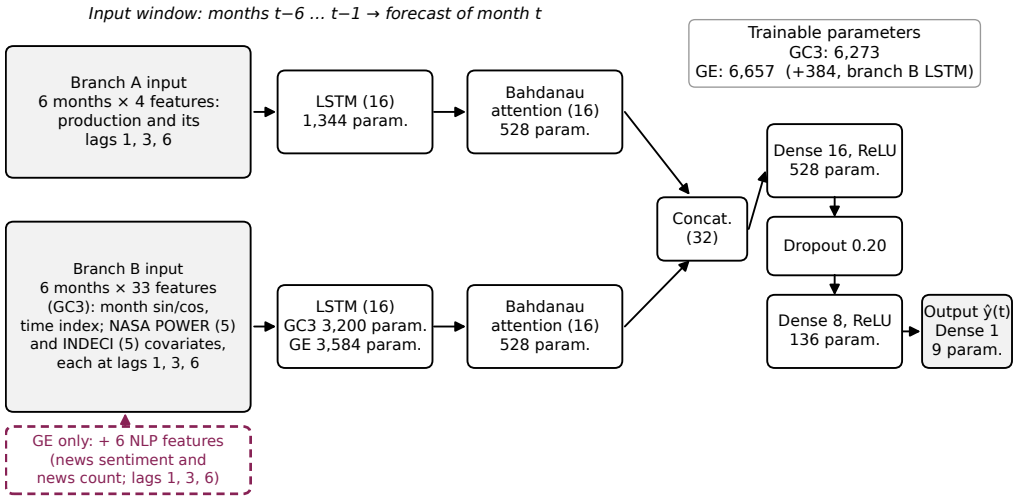


Figure 7. Dual-LSTM with additive (Bahdanau) temporal attention [38] used by GC3 and GE. Each branch receives the previous 6 months: branch A the production series and its lags 1, 3 and 6; branch B calendar terms, a time index and lagged climate (NASA POWER) and disaster (INDECI) covariates. GE adds six lagged news features to branch B (dashed box): the monthly mean sentiment score of agricultural news, computed with the pysentimiento RoBERTuito sentiment model [30,31], and the monthly number of news items. GC3 and GE are otherwise identical. Trainable parameters: GC3 6,273; GE 6,657.

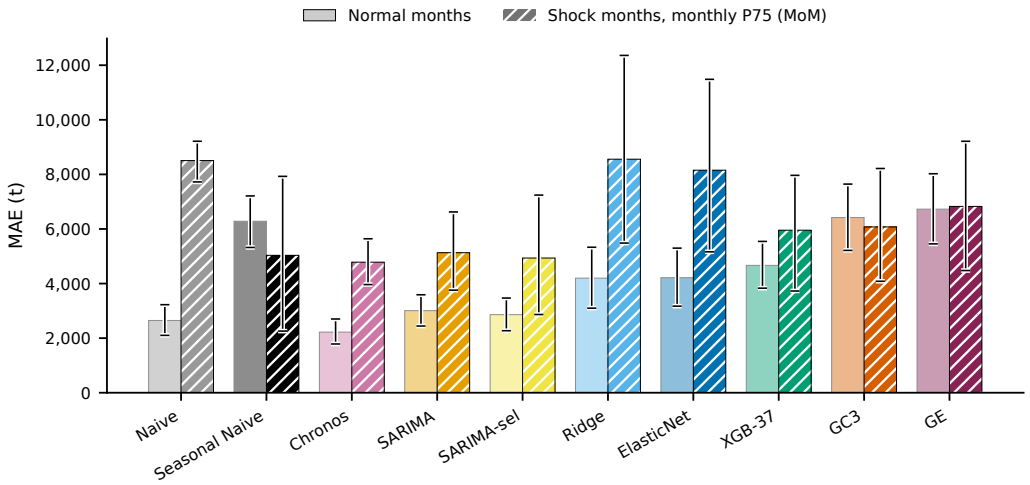


Figure 8. Sour lime

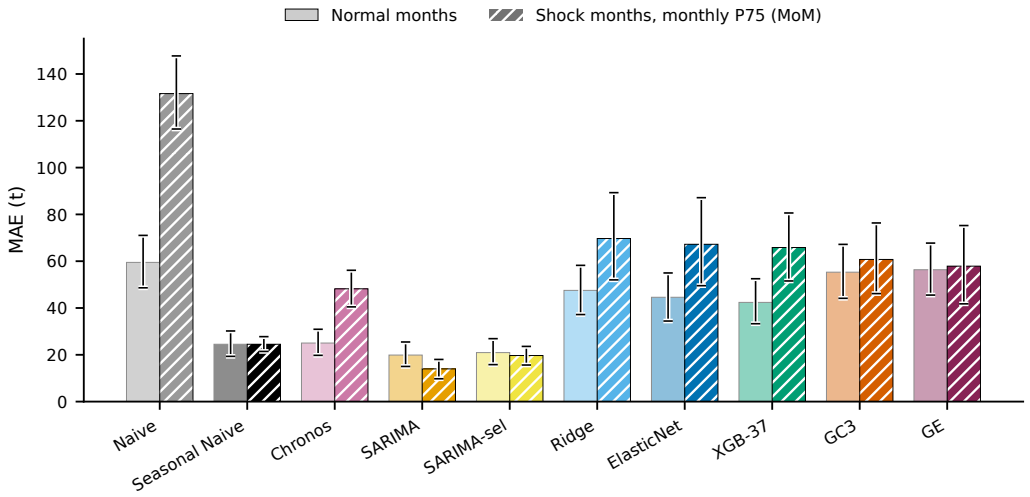


Figure 9. Sweet lime

Figure 10. *Sour lime:* MAE in normal months (light) and in shock months (hatched; monthly P75 threshold) for each model, sour lime (limón sutil), pooled 2021–2025, with paired bootstrap 95% CIs [41]. *Sweet lime:* MAE in normal months (light) and in shock months (hatched; monthly P75 threshold) for each model, sweet lime (limón dulce), pooled 2021–2025, with paired bootstrap 95% CIs.

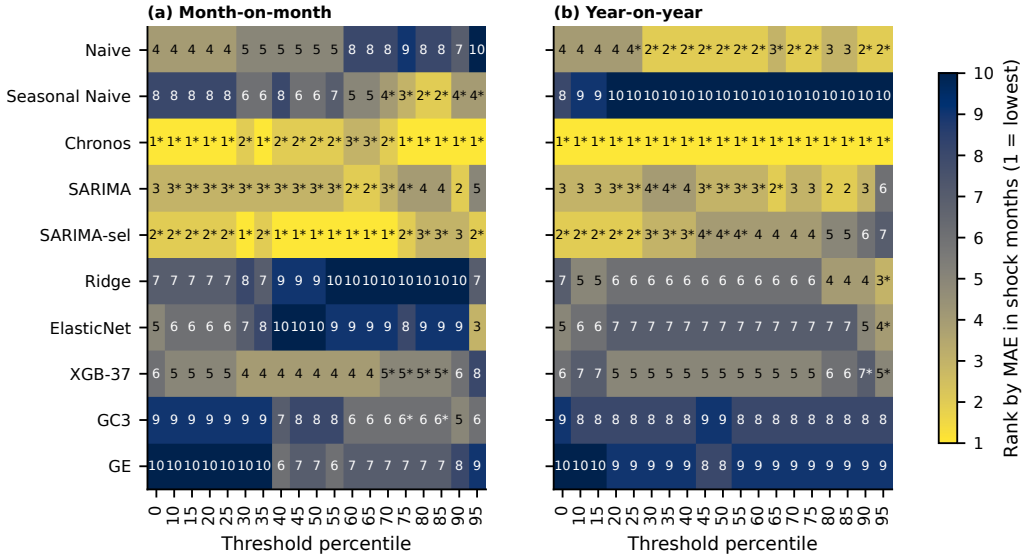


Figure 11. Sour lime

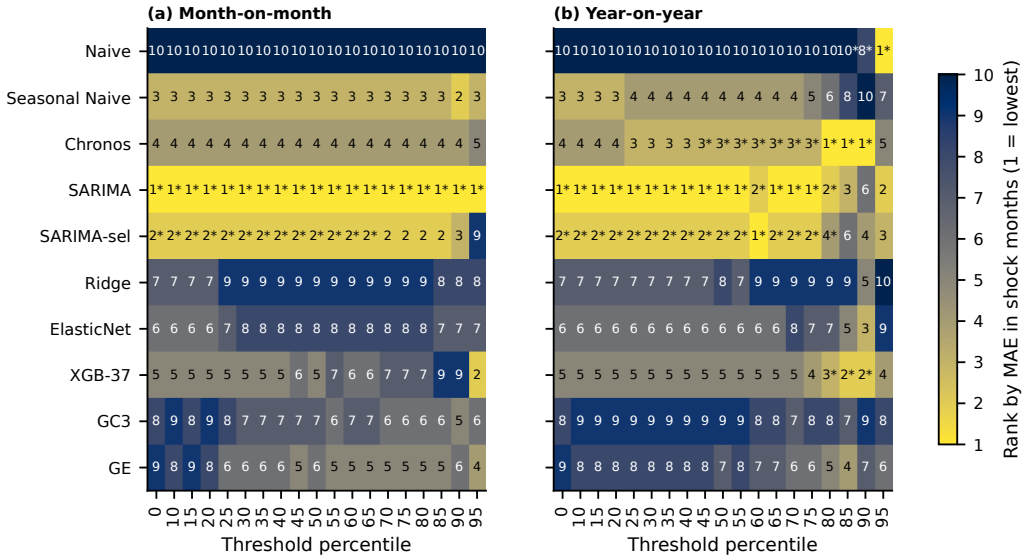


Figure 12. Sweet lime

Figure 13. *Sour lime:* Rank of each model by MAE_shock at every threshold percentile (P0–P95), sour lime (limón sutil), 2021–2025: (a) month-on-month and (b) year-on-year change. Numbers are ranks; * marks the first place and statistical ties with it. *Sweet lime:* Rank of each model by MAE_shock at every threshold percentile (P0–P95), sweet lime (limón dulce), 2021–2025: (a) month-on-month and (b) year-on-year change. Numbers are ranks; * marks the first place and statistical ties with it.

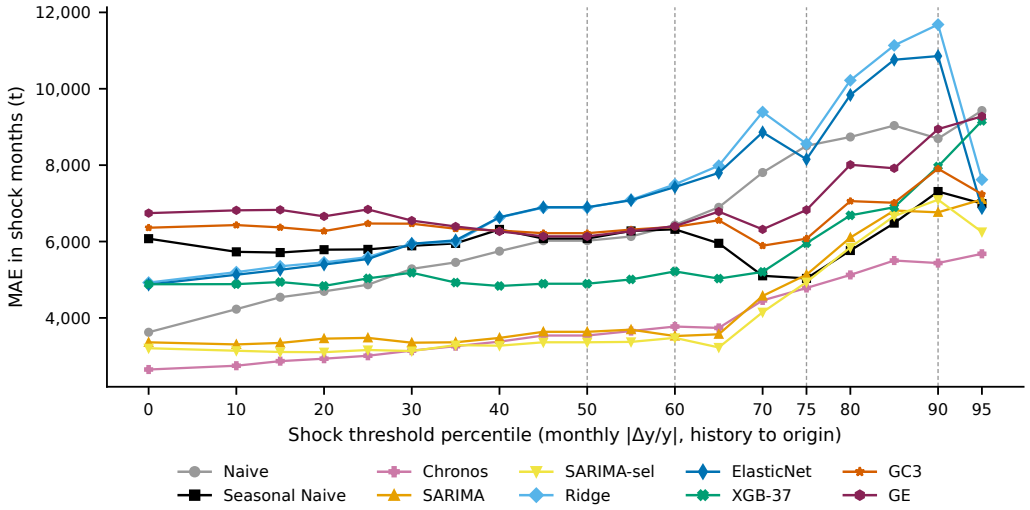


Figure 14. Sour lime

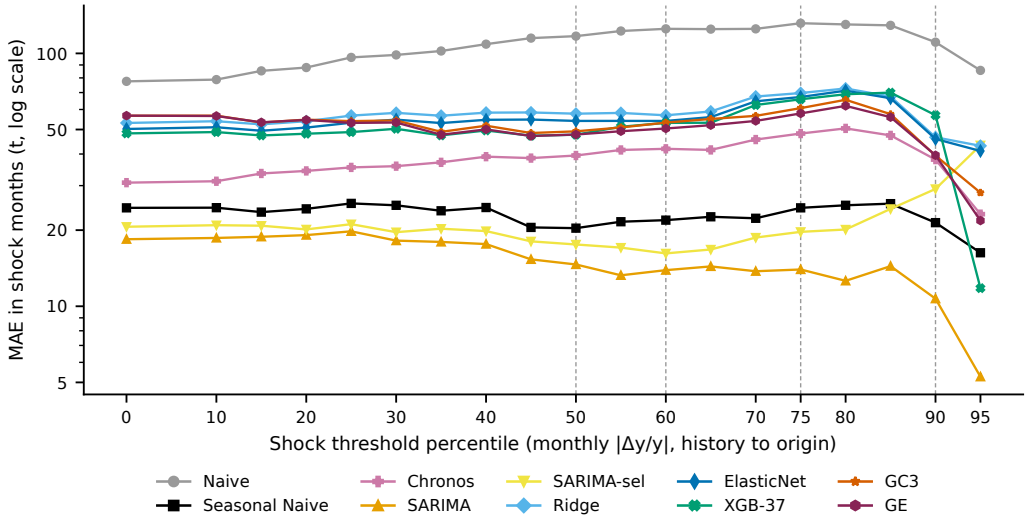


Figure 15. Sweet lime

Figure 16. *Sour lime:* Mean absolute error in shock months (tonnes) versus shock-threshold percentile, sour lime (limón sutil), 2021–2025 pooled. Errors of seeded models are averaged over 10 seeds per month (pre-registered). Dashed lines: P50, P60, P75, P90. *Sweet lime:* Mean absolute error in shock months (tonnes) versus shock-threshold percentile, sweet lime (limón dulce), 2021–2025 pooled. Errors of seeded models are averaged over 10 seeds per month (pre-registered). Dashed lines: P50, P60, P75, P90. Logarithmic y-axis.

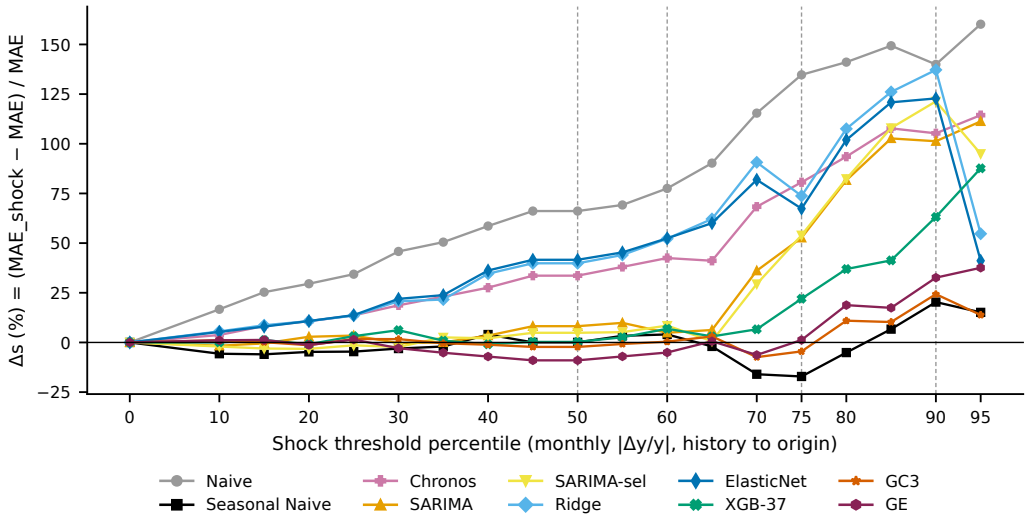


Figure 17. Sour lime

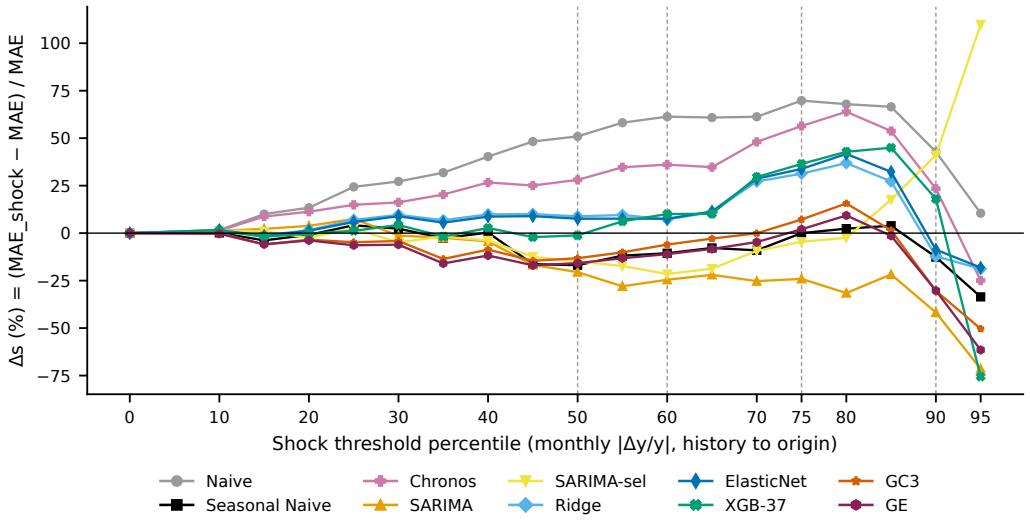


Figure 18. Sweet lime

Figure 19. *Sour lime:* Relative shock penalty Δs versus shock-threshold percentile (P0–P95), sour lime (limón sutil), pooled test windows 2021–2025. Dashed lines mark the reporting points P50, P60, P75 (primary) and P90. *Sweet lime:* Relative shock penalty Δs versus shock-threshold percentile (P0–P95), sweet lime (limón dulce), pooled test windows 2021–2025. Dashed lines mark the reporting points P50, P60, P75 (primary) and P90.

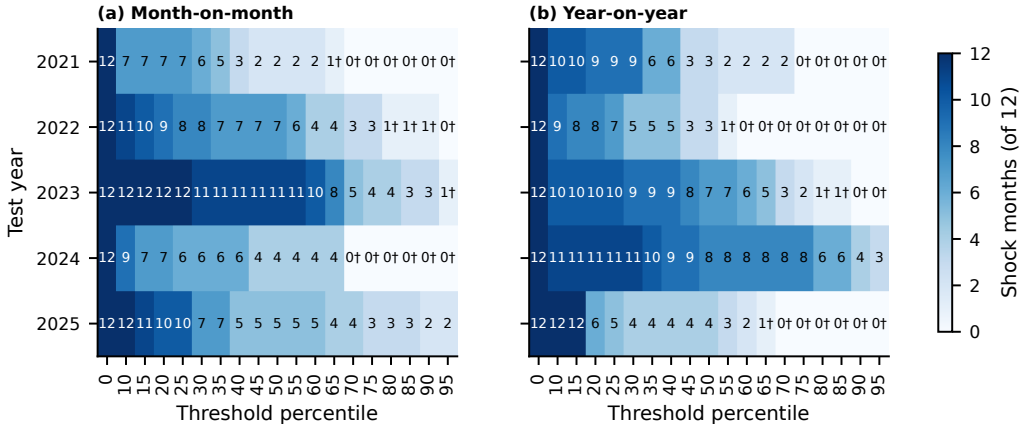


Figure 20. Sour lime

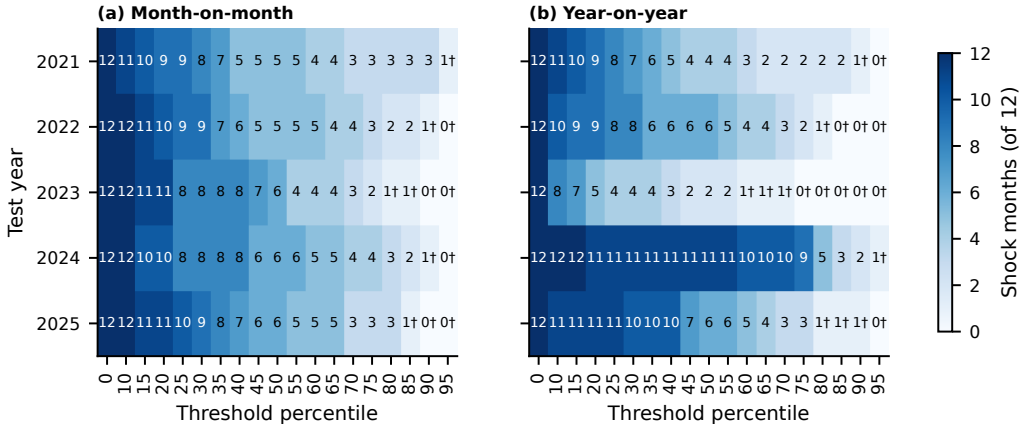


Figure 21. Sweet lime

Figure 22. *Sour lime:* Shock months per test window and threshold percentile, sour lime (limón sutil): (a) month-on-month, (b) year-on-year. Thresholds are computed with the history up to each origin. † = fewer than 2 shocks (non-informative window). *Sweet lime:* Shock months per test window and threshold percentile, sweet lime (limón dulce): (a) month-on-month, (b) year-on-year. Thresholds are computed with the history up to each origin. † = fewer than 2 shocks (non-informative window).

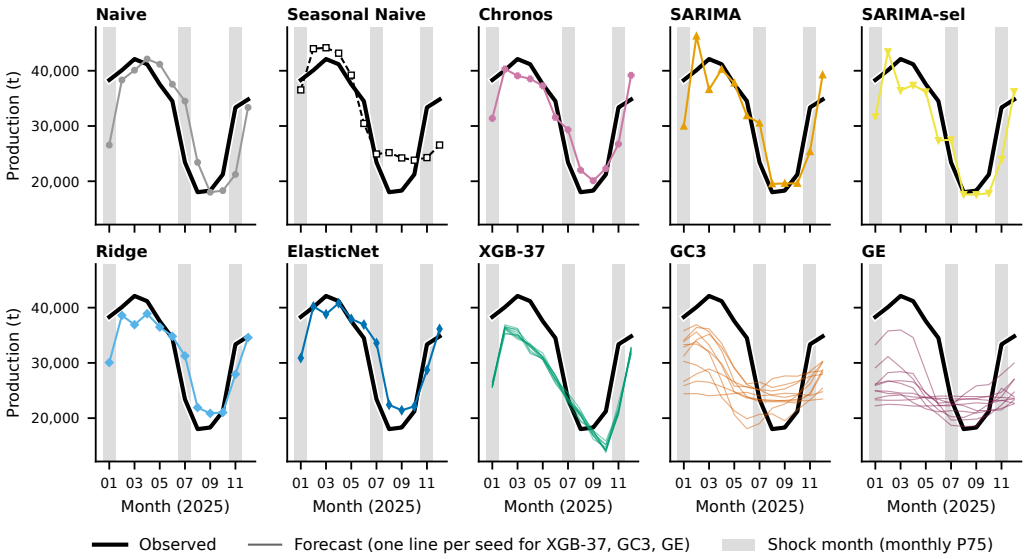


Figure 23. Sour lime

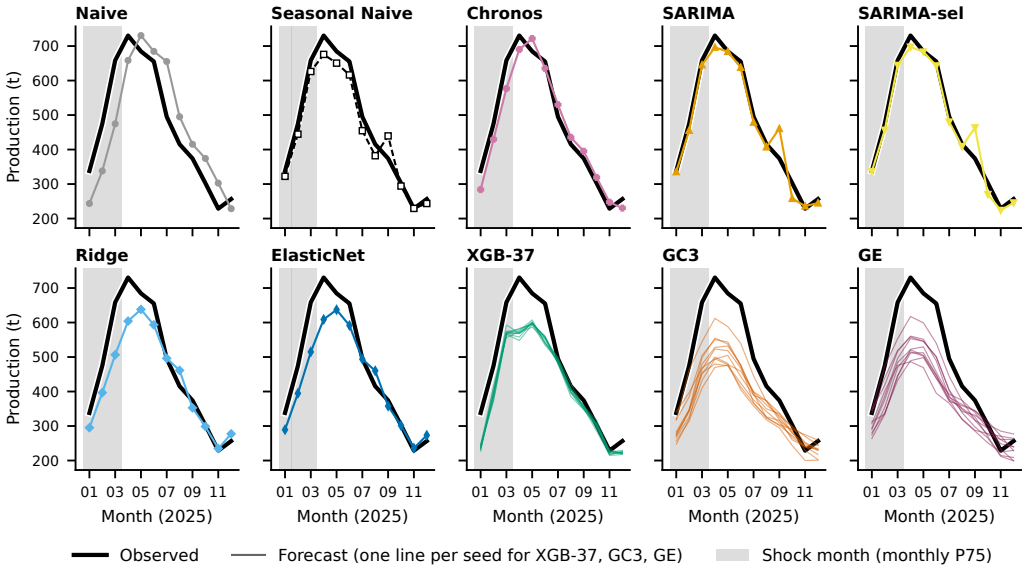


Figure 24. Sweet lime

Figure 25. *Sour lime:* Observed national production and one-step-ahead forecasts of each model, test year 2025, sour lime (limón sutil). Grey: shock months (monthly P75, history to origin). Seeded models: one line per seed (10 seeds). Observed: thick black line; Seasonal Naive: dashed black line with open squares. *Sweet lime:* Observed national production and one-step-ahead forecasts of each model, test year 2025, sweet lime (limón dulce). Grey: shock months (monthly P75, history to origin). Seeded models: one line per seed (10 seeds). Observed: thick black line; Seasonal Naive: dashed black line with open squares.

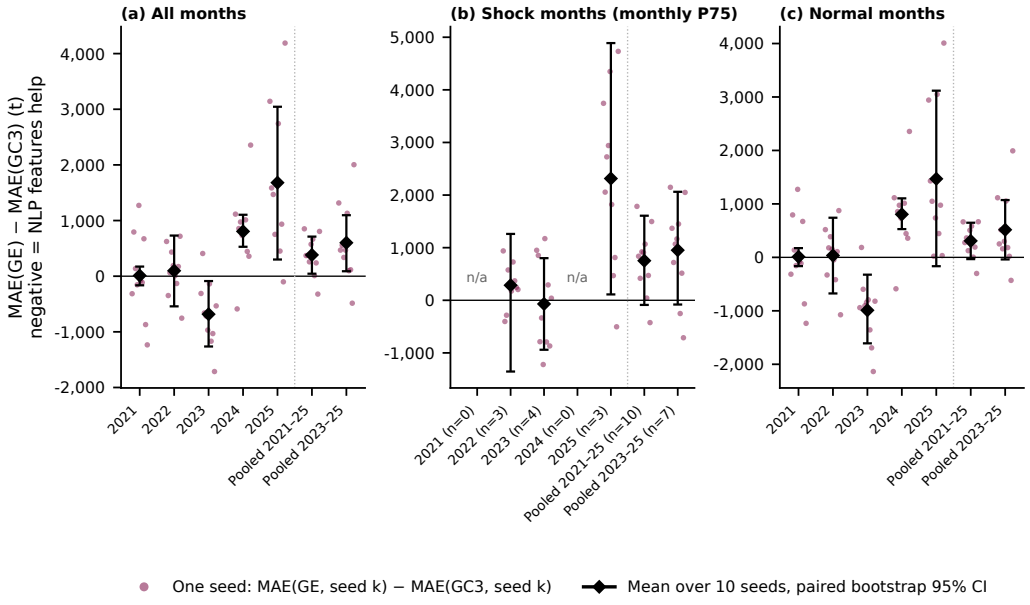


Figure 26. Sour lime

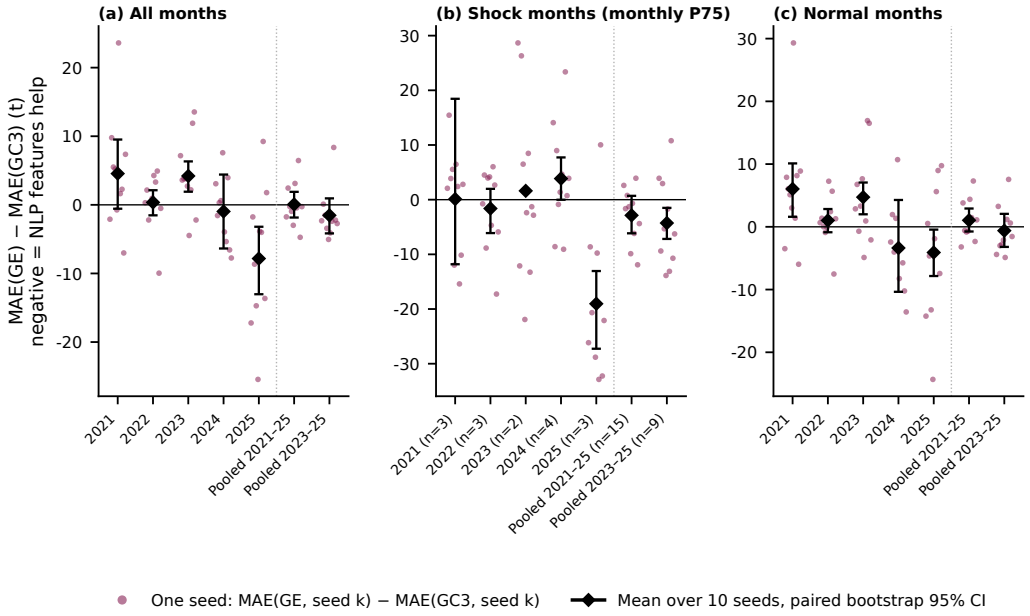


Figure 27. Sweet lime

Figure 28. *Sour lime:* NLP ablation: difference in MAE between GE (Dual-LSTM with NLP features) and GC3 (same architecture without NLP), paired by seed, sour lime (limón sutil), per test window and pooled; (a) all months, (b) shock months (monthly P75), (c) normal months. Dots: individual seed pairs; diamonds: mean with paired bootstrap 95% CI (months resampled within windows, B = 10,000). n/a = no shock month. *Sweet lime:* NLP ablation: difference in MAE between GE (Dual-LSTM with NLP features) and GC3 (same architecture without NLP), paired by seed, sweet lime (limón dulce), per test window and pooled; (a) all months, (b) shock months (monthly P75), (c) normal months. Dots: individual seed pairs; diamonds: mean with paired bootstrap 95% CI (months resampled within windows, B = 10,000). n/a = no shock month.

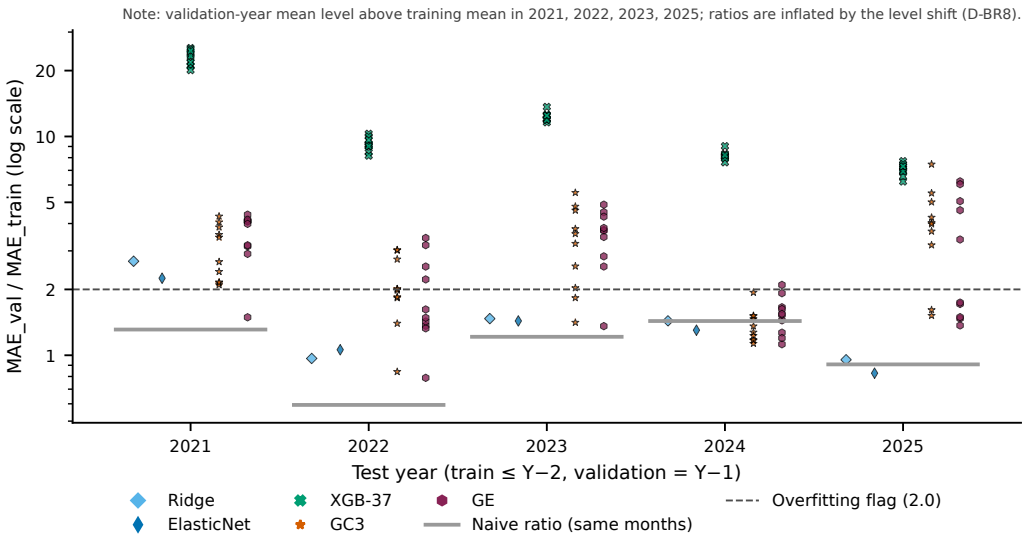


Figure 29. Sour lime

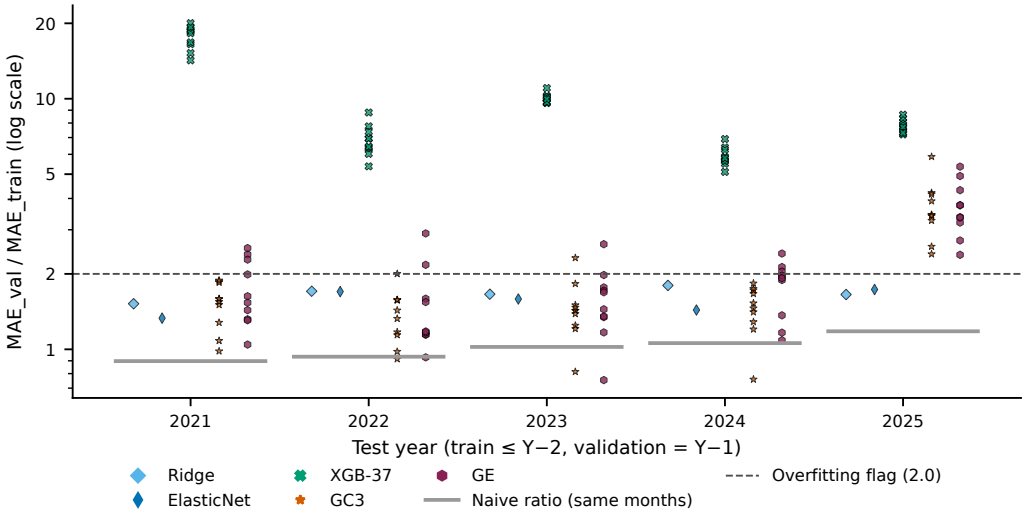


Figure 30. Sweet lime

Figure 31. *Sour lime:* Validation/training MAE ratio for each trained model, window and seed (points), sour lime (limón sutil); grey bars: same ratio for the Naive forecast on the training and validation months of the tabular models (on the LSTM months, which start in 2017-01, it differs by at most 0.030); dashed line: overfitting flag at 2.0 (pre-registered). In Sutil the validation year has a higher mean level than the training period in 2021, 2022, 2023, 2025, which inflates the ratio. *Sweet lime:* Validation/training MAE ratio for each trained model, window and seed (points), sweet lime (limón dulce); grey bars: same ratio for the Naive forecast on the training and validation months of the tabular models (on the LSTM months, which start in 2017-01, it differs by at most 0.015); dashed line: overfitting flag at 2.0 (pre-registered).

Table 5. Lime-producing provinces of Peru, January 2016–December 2025 (MIDAGRI–SISAGRI (accessed [access date])). Shares are relative to the national total of each cu
months with production count the months with production > 0 out of 120.

Department	Province	Latitude	Longitude	Coordinate	Category	Sour lime 2016-2025 (t)	Sour lime share (%)	Months with sour lime > 0 (of 120)	Sweet lime 2016-2025 (t)	Sweet lime share (%)
PIURA	PIURA	-5.15	-80.66	province	Sour lime only	926,666.00	30.58	120.00	0.00	0.00
PIURA	SULLANA	-4.89	-80.69	province	Sour lime only	678,010.00	22.37	120.00	0.00	0.00
LAMBAYEQUE	LAMBAYEQUE	-6.71	-79.90	province	Sour lime only	386,555.00	12.76	120.00	0.00	0.00
TUMBES	ZARUMILLA	-3.50	-80.28	province	Sour lime only	361,442.10	11.93	120.00	0.00	0.00
PIURA	MORROPON	-5.10	-80.16	province	Sour lime only	132,988.00	4.39	120.00	0.00	0.00
TUMBES	TUMBES	-3.57	-80.46	province	Sour lime only	74,185.40	2.45	120.00	0.00	0.00
UCAYALI	CORONEL PORTILLO	-8.37	-74.54	province	Sour lime only	61,919.70	2.04	120.00	0.00	0.00
LORETO	MAYNAS	-3.75	-73.24	province	Both cultivars	55,277.00	1.82	120.00	942.00	1.92
AMAZONAS	UTCUBAMBA	-5.75	-78.44	province	Both cultivars	52,073.80	1.72	120.00	3,150.10	6.43
UCAYALI	PADRE ABAD	-9.03	-75.51	province	Sour lime only	32,399.90	1.07	120.00	0.00	0.00
LORETO	ALTO AMAZONAS	-5.88	-76.13	province	Both cultivars	26,941.00	0.89	120.00	67.00	0.14
JUNIN	SATIPO	-11.25	-74.64	province	Both cultivars	25,734.80	0.85	120.00	14,818.60	30.25
TUMBES	CONTRALMIRANTE VILLAR	-3.68	-80.67	province	Sour lime only	16,441.30	0.54	120.00	0.00	0.00
CUSCO	LA CONVENCION	-12.86	-72.69	province	Sour lime only	16,102.80	0.53	118.00	0.00	0.00
UCAYALI	ATALAYA	-10.73	-73.76	province	Sour lime only	16,010.20	0.53	120.00	0.00	0.00
LORETO	UCAYALI	-7.35	-75.01	province	Sour lime only	10,926.00	0.36	120.00	0.00	0.00
SAN MARTIN	SAN MARTIN	-6.49	-76.36	province	Sour lime only	9,544.10	0.31	120.00	0.00	0.00
SAN MARTIN	PICOTA	-6.92	-76.33	province	Sour lime only	8,634.60	0.28	120.00	0.00	0.00
HUANUCO	LEONCIO PRADO	-9.30	-76.00	province	Sour lime only	8,021.60	0.26	119.00	0.00	0.00
SAN MARTIN	LAMAS	-6.42	-76.52	province	Sour lime only	7,705.30	0.25	120.00	0.00	0.00
JUNIN	CHANCHAMAYO	-11.06	-75.33	province	Both cultivars	7,518.10	0.25	120.00	20,007.60	40.84
ICA	ICA	-14.06	-75.73	province	Sour lime only	6,788.40	0.22	74.00	0.00	0.00
LORETO	LORETO	-4.50	-73.57	province	Sour lime only	5,785.00	0.19	120.00	0.00	0.00
SAN MARTIN	HUALLAGA	-6.94	-76.77	province	Sour lime only	4,744.00	0.16	120.00	0.00	0.00
LORETO	MARISCAL RAMON CASTILLA	-3.91	-70.52	province	Sour lime only	4,463.00	0.15	110.00	0.00	0.00
ANCASH	CASMA	-9.48	-78.31	province	Both cultivars	4,240.00	0.14	120.00	358.00	0.73
LORETO	REQUENA	-5.06	-73.86	province	Both cultivars	3,935.00	0.13	120.00	1,094.00	2.23
ICA	PISCO	-13.71	-76.20	province	Sour lime only	3,862.80	0.13	54.00	0.00	0.00
ICA	NASCA	-14.07	-75.73	department reference	Sour lime only	3,794.00	0.13	31.00	0.00	0.00
MADRE DE DIOS	TAMBOPATA	-12.59	-69.18	province	Sour lime only	3,780.20	0.12	120.00	0.00	0.00
SAN MARTIN	MARISCAL CACERES	-7.18	-76.72	province	Both cultivars	3,678.70	0.12	111.00	199.80	0.41
PIURA	AYABACA	-4.64	-79.72	province	Both cultivars	3,610.90	0.12	113.00	1.00	0.00
LORETO	DATEM DEL MARAÑON	-3.75	-73.25	department reference	Sour lime only	3,254.00	0.11	110.00	0.00	0.00
SAN MARTIN	BELLAVISTA	-7.05	-76.59	province	Sour lime only	3,180.00	0.10	120.00	0.00	0.00
SAN MARTIN	MOYOBAMBA	-6.03	-76.97	province	Sour lime only	2,903.80	0.10	84.00	0.00	0.00

Table 5. Lime-producing provinces of Peru, January 2016–December 2025 (MIDAGRI–SISAGRI (accessed [access date])). Shares are relative to the national total of each cu
months with production count the months with production > 0 out of 120.

Department	Province	Latitude	Longitude	Coordinate	Category	Sour lime 2016-2025 (t)	Sour lime share (%)	Months with sour lime > 0 (of 120)	Sweet lime 2016-2025 (t)	Sweet lime share (%)
SAN MARTIN	TOCACHE	-8.19	-76.51	province	Sour lime only	2,898.80	0.10	109.00	0.00	0.00
AMAZONAS	LUYA	-6.14	-77.95	province	Both cultivars	2,812.70	0.09	120.00	330.90	0.68
AMAZONAS	BAGUA	-5.64	-78.53	province	Both cultivars	2,484.90	0.08	120.00	951.70	1.94
AMAZONAS	CHACHAPOYAS	-6.23	-77.87	province	Sour lime only	2,427.00	0.08	120.00	0.00	0.00
SAN MARTIN	RIOJA	-6.06	-77.17	province	Sour lime only	2,318.80	0.08	109.00	0.00	0.00
PIURA	HUANCABAMBA	-5.24	-79.45	province	Both cultivars	2,031.20	0.07	118.00	30.20	0.06
UCAYALI	PURUS	-9.77	-70.71	province	Sour lime only	2,029.80	0.07	120.00	0.00	0.00
CAJAMARCA	SAN MIGUEL	-7.00	-78.85	province	Sour lime only	1,909.70	0.06	77.00	0.00	0.00
SAN MARTIN	EL DORADO	-6.61	-76.70	province	Sour lime only	1,855.70	0.06	114.00	0.00	0.00
LA LIBERTAD	VIRU	-8.41	-78.75	province	Sour lime only	1,720.90	0.06	120.00	0.00	0.00
ANCASH	HUAYLAS	-9.05	-77.80	province	Sour lime only	1,701.00	0.06	120.00	0.00	0.00
CAJAMARCA	CHOTA	-6.56	-78.65	province	Both cultivars	1,634.00	0.05	92.00	956.00	1.95
HUANUCO	HUAMALIES	-9.55	-76.82	province	Sour lime only	1,543.50	0.05	118.00	0.00	0.00
PIURA	PAITA	-5.09	-81.10	province	Sour lime only	1,444.00	0.05	110.00	0.00	0.00
MADRE DE DIOS	MANU	-12.84	-71.37	province	Sour lime only	1,404.00	0.05	120.00	0.00	0.00
LORETO	PUTUMAYO	-2.45	-72.67	province	Sour lime only	1,398.00	0.05	120.00	0.00	0.00
LAMBAYEQUE	FERREÑAFE	-6.64	-79.79	province	Both cultivars	1,376.80	0.05	54.00	71.00	0.14
APURIMAC	ABANCAY	-13.63	-72.89	province	Sour lime only	1,375.00	0.05	113.00	0.00	0.00
CAJAMARCA	JAEN	-5.71	-78.81	province	Both cultivars	1,253.30	0.04	120.00	1,528.10	3.12
MADRE DE DIOS	TAHUAMANU	-10.95	-69.58	province	Sour lime only	1,240.40	0.04	120.00	0.00	0.00
JUNIN	HUANCAYO	-12.07	-75.21	province	Sour lime only	1,196.70	0.04	120.00	0.00	0.00
AYACUCHO	LA MAR	-13.01	-73.98	province	Sour lime only	1,176.80	0.04	113.00	0.00	0.00
PASCO	OXAPAMPA	-10.57	-75.40	province	Sour lime only	1,030.30	0.03	28.00	0.00	0.00
HUANCAVELICA	TAYACAJA	-12.40	-74.87	province	Both cultivars	952.10	0.03	52.00	170.70	0.35
LA LIBERTAD	BOLIVAR	-7.15	-77.70	province	Sour lime only	799.80	0.03	104.00	0.00	0.00
APURIMAC	CHINCHEROS	-13.52	-73.72	province	Sour lime only	787.40	0.03	43.00	0.00	0.00
APURIMAC	ANDAHUAYLAS	-13.66	-73.39	province	Sour lime only	751.40	0.02	46.00	0.00	0.00
HUANUCO	HUANUCO	-9.93	-76.24	province	Sour lime only	676.60	0.02	115.00	0.00	0.00
LA LIBERTAD	PATAZ	-8.28	-77.30	province	Sour lime only	664.10	0.02	91.00	0.00	0.00
CAJAMARCA	CAJABAMBA	-7.62	-78.05	province	Sour lime only	661.90	0.02	118.00	0.00	0.00
AYACUCHO	HUANTA	-12.94	-74.25	province	Sour lime only	656.50	0.02	83.00	0.00	0.00
ANCASH	SANTA	-9.04	-78.61	province	Both cultivars	634.00	0.02	119.00	910.00	1.86
ANCASH	HUARMEY	-10.07	-78.15	province	Both cultivars	537.00	0.02	115.00	301.00	0.61
AYACUCHO	LUCANAS	-14.69	-74.12	province	Sour lime only	534.00	0.02	54.00	0.00	0.00
HUANUCO	PUERTO INCA	-9.38	-74.97	province	Sour lime only	520.60	0.02	110.00	0.00	0.00

Table 5. Lime-producing provinces of Peru, January 2016–December 2025 (MIDAGRI–SISAGRI (accessed [access date])). Shares are relative to the national total of each cu
months with production count the months with production > 0 out of 120.

Department	Province	Latitude	Longitude	Coordinate	Category	Sour lime 2016-2025 (t)	Sour lime share (%)	Months with sour lime > 0 (of 120)	Sweet lime 2016-2025 (t)	Sweet lime share (%)
HUANUCO	HUACAYBAMBA	-9.04	-76.95	province	Sour lime only	507.20	0.02	46.00	0.00	0.00
LA LIBERTAD	CHEPEN	-7.23	-79.43	province	Sour lime only	499.50	0.02	67.00	0.00	0.00
AYACUCHO	PAUCAR DEL SARA SARA	-15.28	-73.34	province	Sour lime only	496.00	0.02	44.00	0.00	0.00
AYACUCHO	VILCAS HUAMAN	-13.65	-73.95	province	Sour lime only	479.60	0.02	69.00	0.00	0.00
LA LIBERTAD	SANCHEZ CARRION	-7.81	-78.05	province	Sour lime only	465.80	0.02	108.00	0.00	0.00
PIURA	SECHURA	-5.56	-80.82	province	Sour lime only	464.00	0.02	109.00	0.00	0.00
PUNO	SANDIA	-14.32	-69.47	province	Sour lime only	459.70	0.02	51.00	0.00	0.00
HUANUCO	MARAÑON	-9.93	-76.24	department reference	Sour lime only	458.50	0.02	80.00	0.00	0.00
CAJAMARCA	SAN IGNACIO	-5.15	-79.00	province	Both cultivars	425.70	0.01	76.00	1,274.50	2.60
MOQUEGUA	GENERAL SANCHEZ CERRO	-16.67	-70.97	province	Sour lime only	394.40	0.01	37.00	0.00	0.00
MOQUEGUA	MARISCAL NIETO	-17.19	-70.93	province	Sour lime only	359.70	0.01	35.00	0.00	0.00
HUANCAVELICA	CHURCAMPA	-12.74	-74.39	province	Both cultivars	336.40	0.01	60.00	141.90	0.29
CAJAMARCA	SAN MARCOS	-7.34	-78.17	province	Sour lime only	324.90	0.01	108.00	0.00	0.00
LIMA	HUAURA	-11.11	-77.61	province	Sour lime only	308.90	0.01	20.00	0.00	0.00
CAJAMARCA	SANTA CRUZ	-6.63	-78.94	province	Sour lime only	305.00	0.01	50.00	0.00	0.00
AREQUIPA	LA UNION	-15.21	-72.89	province	Sour lime only	281.30	0.01	10.00	0.00	0.00
ICA	PALPA	-14.53	-75.19	province	Sour lime only	258.90	0.01	25.00	0.00	0.00
HUANCAVELICA	ACOBAMBA	-12.84	-74.57	province	Sour lime only	229.60	0.01	51.00	0.00	0.00
APURIMAC	AYMARAES	-14.29	-73.24	province	Sour lime only	189.50	0.01	117.00	0.00	0.00
AYACUCHO	SUCRE	-14.01	-73.84	province	Sour lime only	177.00	0.01	22.00	0.00	0.00
AYACUCHO	HUAMANGA	-13.16	-74.23	province	Sour lime only	175.20	0.01	118.00	0.00	0.00
AREQUIPA	CARAVELI	-15.77	-73.37	province	Sour lime only	165.90	0.01	17.00	0.00	0.00
HUANCAVELICA	ANGARAES	-12.98	-74.72	province	Sour lime only	158.90	0.01	50.00	0.00	0.00
HUANCAVELICA	HUAYTARA	-13.60	-75.35	province	Sour lime only	150.90	0.00	66.00	0.00	0.00
LA LIBERTAD	GRAN CHIMU	-7.48	-78.82	province	Sour lime only	143.90	0.00	70.00	0.00	0.00
AYACUCHO	PARINACOCHAS	-15.02	-73.78	province	Sour lime only	131.00	0.00	40.00	0.00	0.00
HUANCAVELICA	CASTROVIRREYNA	-13.28	-75.32	province	Sour lime only	127.70	0.00	79.00	0.00	0.00
LA LIBERTAD	ASCOPE	-8.11	-79.03	department reference	Sour lime only	121.70	0.00	52.00	0.00	0.00
PUNO	CARABAYA	-14.07	-70.43	province	Sour lime only	107.00	0.00	40.00	0.00	0.00
LA LIBERTAD	OTUZCO	-7.90	-78.57	province	Sour lime only	106.90	0.00	73.00	0.00	0.00
APURIMAC	GRAU	-14.11	-72.71	province	Sour lime only	80.00	0.00	51.00	0.00	0.00
ICA	CHINCHA	-13.42	-76.13	province	Sour lime only	76.40	0.00	13.00	0.00	0.00
CAJAMARCA	CUTERVO	-6.38	-78.82	province	Both cultivars	21.00	0.00	4.00	309.00	0.63
ANCASH	CARHUAZ	-9.53	-77.53	department reference	Sweet lime only	0.00	0.00	0.00	753.00	1.54
ANCASH	YUNGAY	-9.53	-77.53	department reference	Sweet lime only	0.00	0.00	0.00	628.00	1.28

https://doi.org/10.3390/forecast1010000

Table 5. Lime-producing provinces of Peru, January 2016–December 2025 (MIDAGRI–SISAGRI (accessed [access date])). Shares are relative to the national total of each cu
months with production count the months with production > 0 out of 120.

Department	Province	Latitude	Longitude	Coordinate	Category	Sour lime 2016-2025 (t)	Sour lime share (%)	Months with sour lime > 0 (of 120)	Sweet lime 2016-2025 (t)	Sweet lime share (%)
LAMBAYEQUE	CHICLAYO	-6.77	-79.84	department reference	No harvest recorded	0.00	0.00	0.00	0.00	0.00
LIMA	CAÑETE	-12.05	-77.04	department reference	No harvest recorded	0.00	0.00	0.00	0.00	0.00
LIMA	LIMA	-12.05	-77.04	department reference	No harvest recorded	0.00	0.00	0.00	0.00	0.00
PIURA	TALARA	-5.19	-80.63	department reference	No harvest recorded	0.00	0.00	0.00	0.00	0.00
TOTAL						3,030,543.60	100.00		48,994.10	100.00

https://doi.org/10.3390/forecast1010000

Table 6. EXPLORATORY (post-hoc): XGB-43 versus XGB-37, 2021-2025, monthly P75 (t)

Cultivar	Scope	Months	MAE XGB-37	MAE XGB-43	XGB-43 − XGB-37	95% CI lower
sutil	global	60	4,881.4	5,011.9	130.5	1.5
sutil	shock_P75	10	5,955.2	5,997.2	42.0	-240.1
sutil	normal_P75	50	4,666.7	4,814.9	148.2	6.5
dulce	global	60	48.3	47.6	-0.6	-2.4
dulce	shock_P75	15	65.8	62.2	-3.6	-5.9
dulce	normal_P75	45	42.4	42.8	0.4	-1.7

Table 7. MAE por ventana, P75 mensual, Sutil (t)

modelo	2021	2022	2023	2024	2025
Naive	1,986	3,762	4,590	3,078	4,704
Seasonal Naive	4,590	4,136	7,638	9,850	4,160
Chronos	2,541	2,205	3,157	2,023	3,316
SARIMA	3,208	3,061	3,693	2,858	3,985
SARIMA-sel	2,304	2,609	3,836	3,333	3,947
Ridge	3,108	3,187	10,778	4,322	3,232
ElasticNet	3,006	2,968	10,413	4,751	3,223
XGB-37	5,211	4,552	3,785	4,957	5,902
GC3	3,856	5,162	6,074	8,969	7,746
GE	3,867	5,260	5,391	9,774	9,426
XGB-43	5,333	4,664	3,679	5,466	5,918

Table 8. MAE por ventana, P75 mensual, Dulce (t)

modelo	2021	2022	2023	2024	2025
Naive	67.8	73.3	76.2	85.6	84.7
Seasonal Naive	18.4	19.9	11.8	41.8	30.5
Chronos	19.2	30.0	30.4	39.9	34.6
SARIMA	13.7	12.4	18.8	25.5	21.8
SARIMA-sel	27.9	12.4	17.4	25.6	19.8
Ridge	37.9	35.5	43.6	97.7	50.6
ElasticNet	37.6	35.2	43.4	85.7	49.3
XGB-37	25.7	52.3	45.8	58.0	59.4
GC3	36.5	37.2	38.0	68.2	103.4
GE	41.1	37.5	42.2	67.3	95.6
XGB-43	26.2	52.2	44.3	54.1	61.4

Table 9. MAE_shock por ventana, P75 mensual, Sutil (t)

modelo	2021 (n=0)	2022 (n=3)	2023 (n=4)	2024 (n=0)	2025 (n=3)
Naive	–	8,522	6,129	–	11,660
Seasonal Naive	–	4,472	6,135	–	4,130
Chronos	–	4,852	3,454	–	6,487
SARIMA	–	3,905	4,030	–	7,824
SARIMA-sel	–	4,207	4,158	–	6,701
Ridge	–	4,394	12,714	–	7,177
ElasticNet	–	4,015	11,806	–	7,423
XGB-37	–	5,234	4,740	–	8,297
GC3	–	4,916	7,304	–	5,594
GE	–	5,202	7,234	–	7,908
XGB-43	–	5,374	4,656	–	8,408

Table 10. MAE_shock por ventana, P75 mensual, Dulce (t)

modelo	2021 (n=3)	2022 (n=3)	2023 (n=2)	2024 (n=4)	2025 (n=3)
Naive	120.6	123.0	140.7	136.8	138.3
Seasonal Naive	16.0	31.5	7.4	33.2	25.9
Chronos	26.1	48.9	43.8	57.5	60.1
SARIMA	4.3	12.9	8.8	26.4	11.6
SARIMA-sel	34.9	12.9	7.5	25.9	11.0
Ridge	41.9	54.1	29.3	106.5	90.9
ElasticNet	43.5	53.5	29.7	96.5	90.7
XGB-37	42.2	79.6	72.2	51.9	90.1
GC3	43.7	44.3	50.9	47.1	118.9
GE	43.8	42.7	52.5	51.0	99.9
XGB-43	40.0	78.9	73.3	45.9	82.3

Table 11. DM-HLN, p-valor sin ajustar (principal_2021_2025, Sutil). Bajo la diagonal: global (n=60); sobre la diagonal: solo shocks P75 mensual (n=10). Signo: + = el modelo de la fila tiene más error que el de la columna

modelo	Nv	SN	Chr	SAR	SSel	Rdg	EN	X37	GC3	GE
Naive	–	0.094+	0.000+	0.001+	0.013+	0.984-	0.881+	0.045+	0.123+	0.2
Seasonal Naive	0.001+	–	0.867+	0.949-	0.932+	0.235-	0.247-	0.614-	0.331-	0.2
Chronos	0.003-	0.000-	–	0.509-	0.867-	0.145-	0.146-	0.292-	0.255-	0.1
SARIMA	0.548-	0.000-	0.017+	–	0.807+	0.152-	0.151-	0.460-	0.427-	0.1
SARIMA-sel	0.385-	0.000-	0.102+	0.587-	–	0.156-	0.157-	0.358-	0.243-	0.0
Ridge	0.071+	0.180-	0.002+	0.030+	0.022+	–	0.406+	0.333+	0.344+	0.5
ElasticNet	0.066+	0.129-	0.002+	0.028+	0.018+	0.696-	–	0.389+	0.382+	0.5
XGB-37	0.018+	0.063-	0.000+	0.004+	0.001+	0.955-	0.990+	–	0.919-	0.3
GC3	0.000+	0.641+	0.000+	0.000+	0.000+	0.080+	0.050+	0.001+	–	0.2
GE	0.000+	0.341+	0.000+	0.000+	0.000+	0.045+	0.030+	0.000+	0.061+	–

Table 12. DM-HLN, p-valor sin ajustar (principal_2021_2025, Dulce). Bajo la diagonal: global (n=60); sobre la diagonal: solo shocks P75 mensual (n=15). Signo: + = el modelo de la fila tiene más error que el de la columna

modelo	Nv	SN	Chr	SAR	SSel	Rdg	EN	X37	GC3	GE
Naive	–	0.000+	0.000+	0.000+	0.000+	0.000+	0.000+	0.000+	0.000+	0.00
Seasonal Naive	0.000-	–	0.000-	0.001+	0.237+	0.001-	0.001-	0.001-	0.004-	0.00
Chronos	0.000-	0.054+	–	0.000+	0.001+	0.031-	0.038-	0.024-	0.183-	0.23
SARIMA	0.000-	0.004-	0.000-	–	0.115-	0.000-	0.000-	0.000-	0.001-	0.00
SARIMA-sel	0.000-	0.121-	0.005-	0.203+	–	0.001-	0.001-	0.001-	0.005-	0.00
Ridge	0.001-	0.000+	0.000+	0.000+	0.000+	–	0.270+	0.767+	0.387+	0.20
ElasticNet	0.000-	0.000+	0.000+	0.000+	0.000+	0.009-	–	0.907+	0.496+	0.27
XGB-37	0.000-	0.000+	0.000+	0.000+	0.000+	0.377-	0.693-	–	0.644+	0.44
GC3	0.013-	0.000+	0.000+	0.000+	0.000+	0.533+	0.242+	0.075+	–	0.34
GE	0.012-	0.000+	0.000+	0.000+	0.000+	0.508+	0.216+	0.061+	0.954+	–

Table 13. Ablación NLP: MAE(GE) - MAE(GC3) emparejado por semilla, Sutil (t; negativo = GE mejor)

conjunto	alcance	n meses	GE-GC3 (t)	IC95	DE semillas	semillas
2021	global	12	11	[-164; 173]	756	6
2021	shock_P75	0	–	–	–	–
2021	normal_P75	12	11	[-164; 173]	756	6
2022	global	12	98	[-542; 730]	444	3
2022	shock_P75	3	286	[-1,354; 1,261]	413	2
2022	normal_P75	9	35	[-675; 741]	545	4
2023	global	12	-683	[-1,263; -88]	592	9
2023	shock_P75	4	-70	[-940; 801]	859	5
2023	normal_P75	8	-990	[-1,608; -324]	630	9
2024	global	12	805	[529; 1,104]	734	1
2024	shock_P75	0	–	–	–	–
2024	normal_P75	12	805	[529; 1,104]	734	1
2025	global	12	1,680	[300; 3,045]	1,322	1
2025	shock_P75	3	2,314	[112; 4,889]	1,716	1
2025	normal_P75	9	1,469	[-165; 3,118]	1,386	0
principal_2021_2025	global	60	382	[45; 711]	363	1
principal_2021_2025	shock_P75	10	752	[-88; 1,607]	655	1
principal_2021_2025	normal_P75	50	308	[-33; 648]	314	1
sensibilidad_2023_2025	global	36	601	[89; 1,096]	710	1
sensibilidad_2023_2025	shock_P75	7	952	[-81; 2,062]	918	2
sensibilidad_2023_2025	normal_P75	29	516	[-39; 1,072]	693	1

Table 14. Ablación NLP: MAE(GE) - MAE(GC3) emparejado por semilla, Dulce (t; negativo = GE mejor)

conjunto	alcance	n meses	GE-GC3 (t)	IC95	DE semillas	semillas G
2021	global	12	4.6	[-0.6; 9.5]	8.3	3
2021	shock_P75	3	0.1	[-11.8; 18.4]	9.6	3
2021	normal_P75	9	6.0	[1.6; 10.1]	9.6	2
2022	global	12	0.4	[-1.5; 2.1]	4.2	3
2022	shock_P75	3	-1.6	[-6.1; 2.0]	7.5	5
2022	normal_P75	9	1.0	[-0.9; 2.8]	4.0	3
2023	global	12	4.2	[1.9; 6.3]	5.6	2
2023	shock_P75	2	1.6	[1.5; 1.7]	16.4	6
2023	normal_P75	10	4.7	[2.0; 7.1]	7.4	3
2024	global	12	-1.0	[-6.4; 4.4]	5.0	5
2024	shock_P75	4	3.9	[-0.0; 7.7]	9.8	3
2024	normal_P75	8	-3.4	[-10.4; 4.3]	6.9	7
2025	global	12	-7.8	[-13.0; -3.2]	10.2	8
2025	shock_P75	3	-19.0	[-27.3; -13.1]	13.2	9
2025	normal_P75	9	-4.1	[-7.8; -0.4]	11.0	6
principal_2021_2025	global	60	0.1	[-1.9; 1.9]	3.2	7
principal_2021_2025	shock_P75	15	-2.9	[-6.2; 0.7]	5.2	7
principal_2021_2025	normal_P75	45	1.0	[-0.7; 2.9]	3.3	5
sensibilidad_2023_2025	global	36	-1.5	[-4.2; 0.9]	3.7	8
sensibilidad_2023_2025	shock_P75	9	-4.3	[-7.2; -1.5]	8.1	7
sensibilidad_2023_2025	normal_P75	27	-0.6	[-3.2; 2.1]	3.8	6

4. Discussion

Parsimony and pretrained benchmarks. The results do not support the idea that greater model capacity yields lower error when the training set is small. In sour lime the lowest errors came from Chronos, applied without local training, and from SARIMA variants; in sweet lime from SARIMA. The DualLSTM variants had the highest error in sour lime and were outperformed in sweet lime by the statistical models, the Seasonal

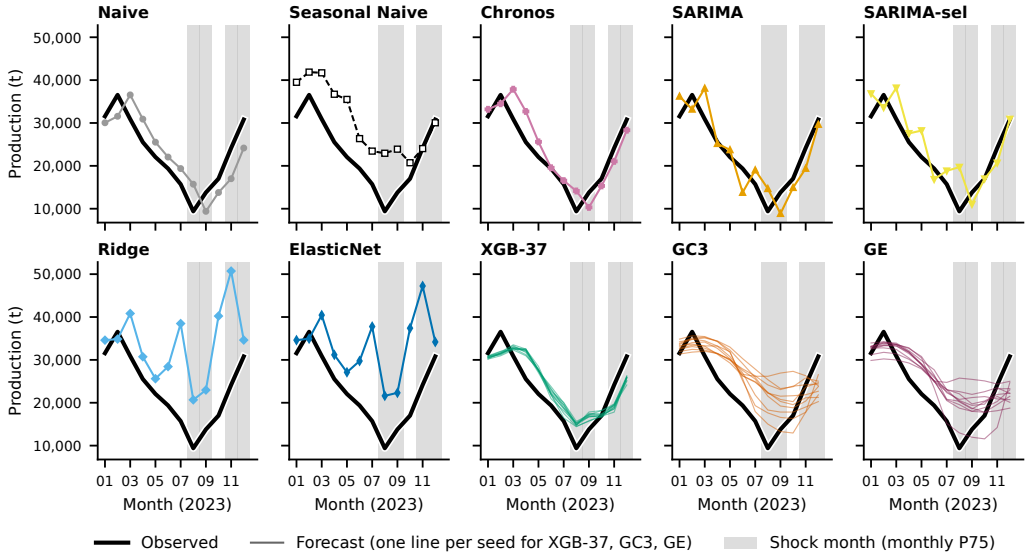


Figure 32. Sour lime, 2023

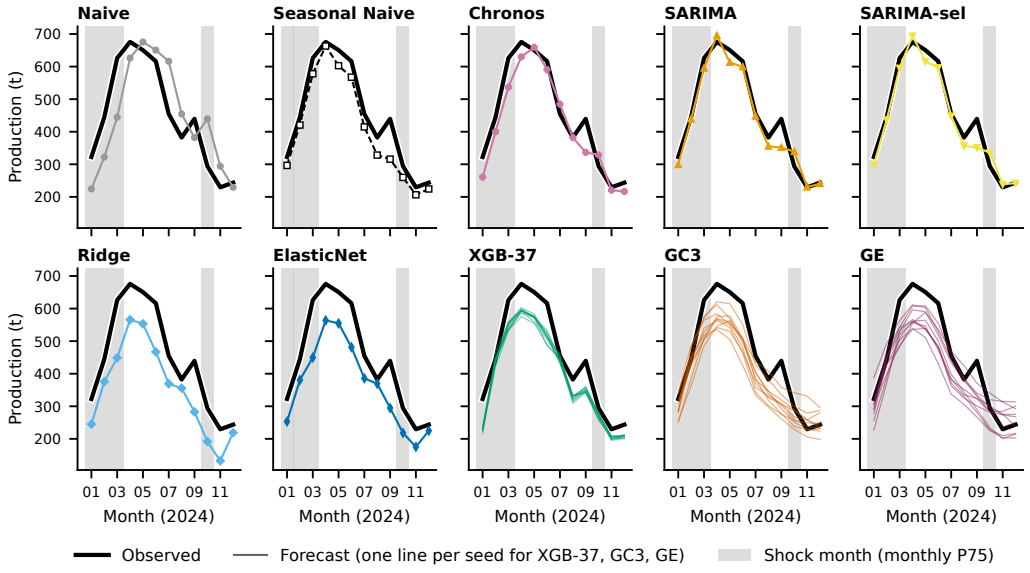


Figure 33. Sweet lime, 2024

Figure 34. *Sour lime, 2023:* Observed national production and one-step-ahead forecasts of each model, test year 2023, sour lime (limón sutil). Grey: shock months (monthly P75, history to origin). Seeded models: one line per seed (10 seeds). Observed: thick black line; Seasonal Naive: dashed black line with open squares. *Sweet lime, 2024:* Observed national production and one-step-ahead forecasts of each model, test year 2024, sweet lime (limón dulce). Grey: shock months (monthly P75, history to origin). Seeded models: one line per seed (10 seeds). Observed: thick black line; Seasonal Naive: dashed black line with open squares.

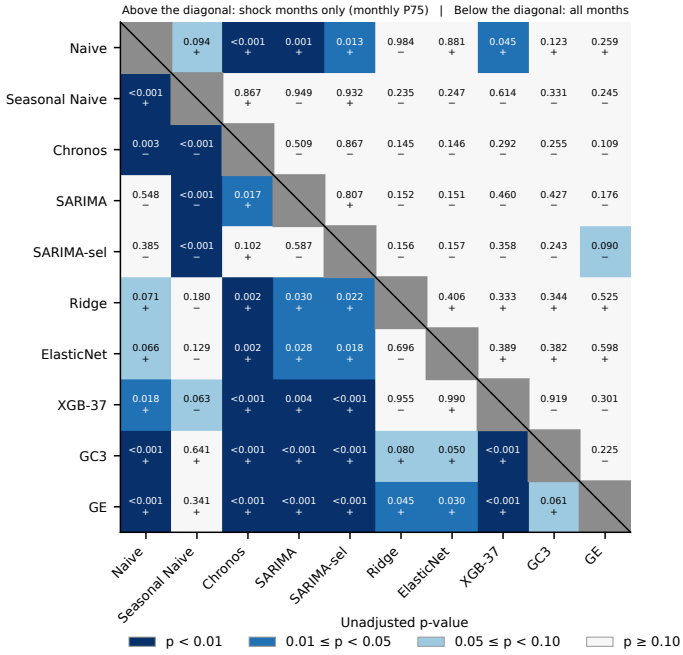


Figure 35. Sour lime

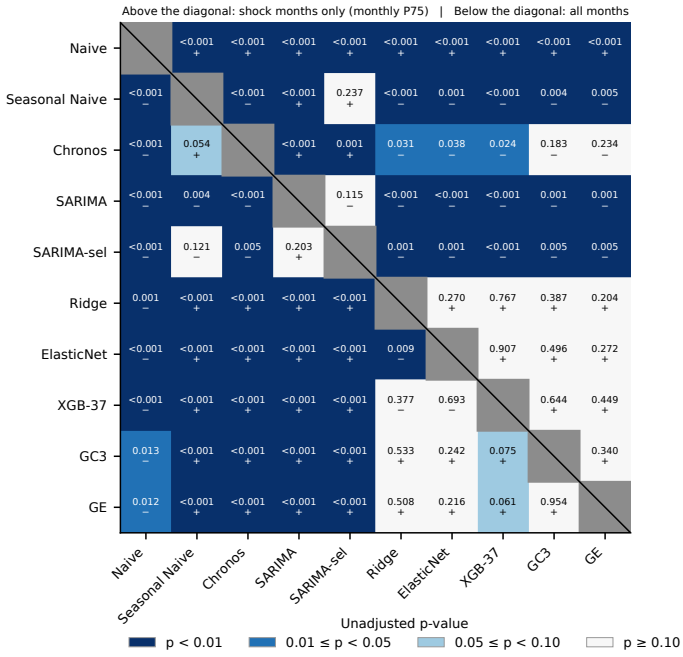


Figure 36. Sweet lime

Figure 37. *Sour lime:* Diebold–Mariano tests [42] with the Harvey–Leybourne–Newbold correction [43] (h = 1, absolute loss, seed-averaged errors, pooled windows 2021–2025), sour lime (limón sutil): unadjusted p-values. Lower triangle: all test months (n = 60); upper triangle: shock months only (monthly P75, n = 10). '+' = row model has larger error than column model. *Sweet lime:* Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction (h = 1, absolute loss, seed-averaged errors, pooled windows 2021–2025), sweet lime (limón dulce): unadjusted p-values. Lower triangle: all test months (n = 60); upper triangle: shock months only (monthly P75, n = 15). '+' = row model has larger error than column model.

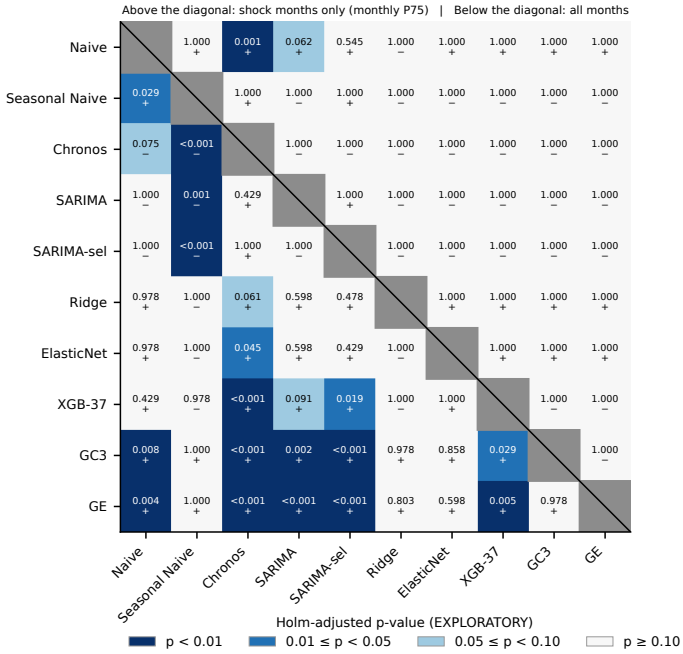


Figure 38. Sour lime

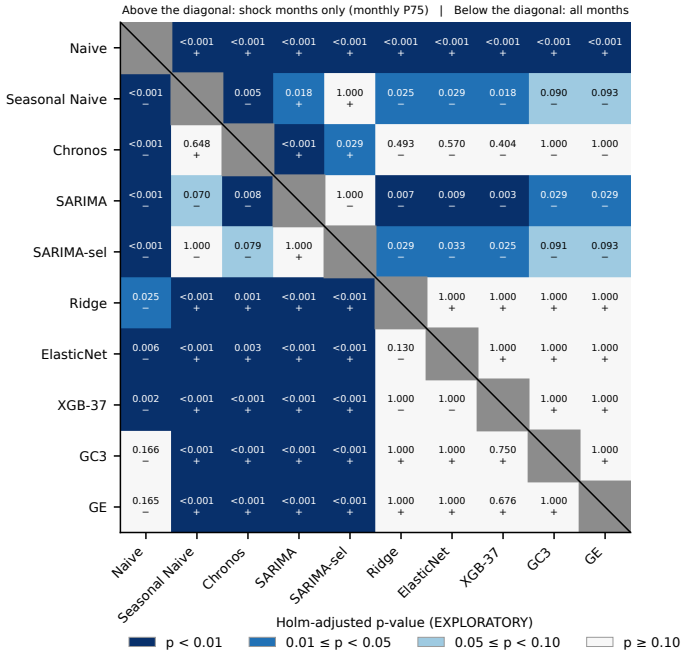


Figure 39. Sweet lime

Figure 40. EXPLORATORY. *Sour lime:* Diebold–Mariano tests [42] with the Harvey–Leybourne–Newbold correction [43] (h = 1, absolute loss, seed-averaged errors, pooled windows 2021–2025), sour lime (limón sutil): Holm-adjusted [44] (exploratory; family = 45 pairs per triangle) p-values. Lower triangle: all test months (n = 60); upper triangle: shock months only (monthly P75, n = 10). '+' = row model has larger error than column model. Holm correction is an exploratory addition, not part of the pre-registration. *Sweet lime:* Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction (h = 1, absolute loss, seed-averaged errors, pooled windows 2021–2025), sweet lime (limón dulce): Holm-adjusted (exploratory; family = 45 pairs per triangle) p-values. Lower triangle: all test months (n = 60); upper triangle: shock months only (monthly P75, n = 15). '+' = row model has larger error than column model. Holm correction is an exploratory addition, not part of the pre-registration.

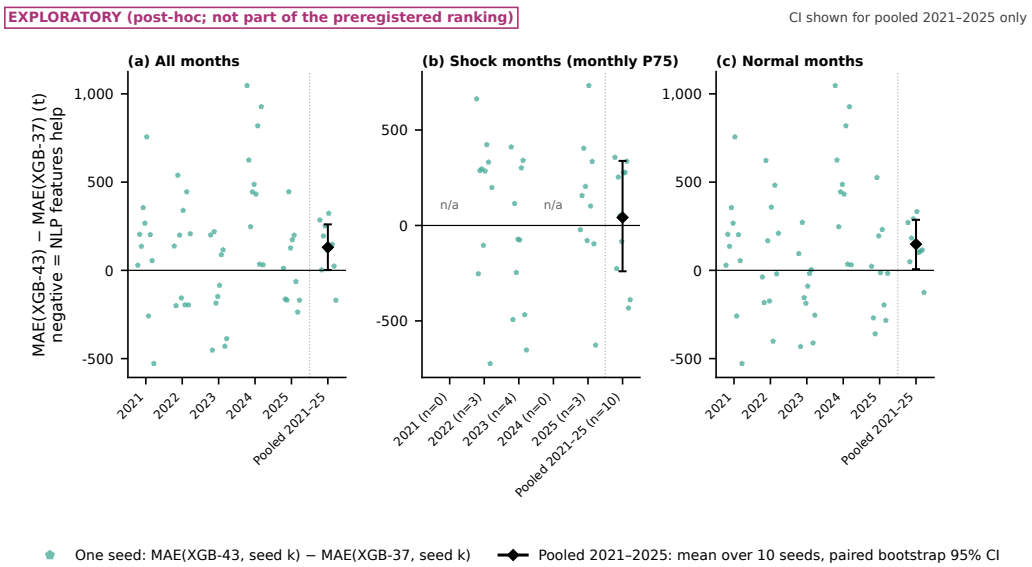


Figure 41. Sour lime

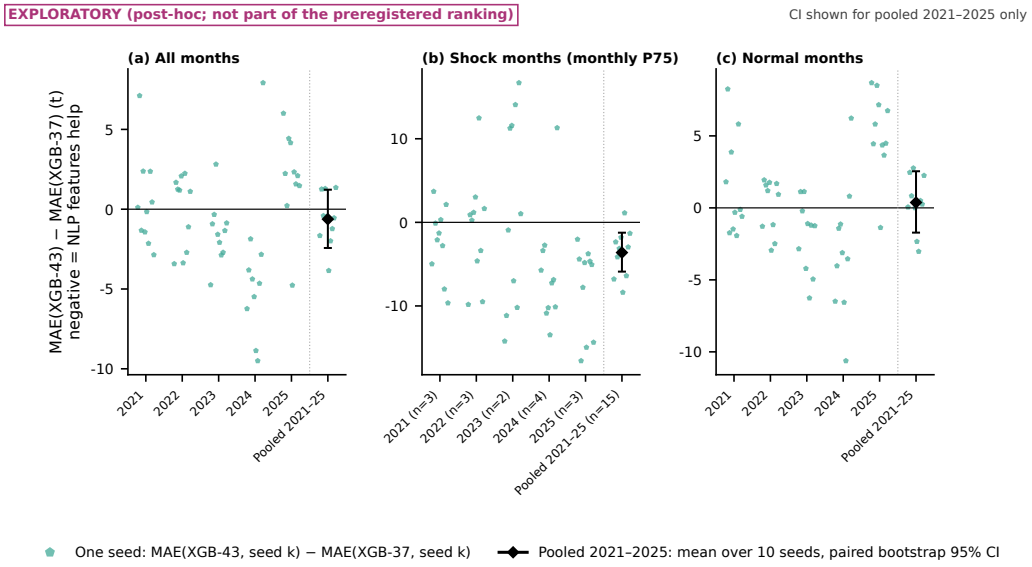


Figure 42. Sweet lime

Figure 43. *Sour lime:* EXPLORATORY (post-hoc) comparison: difference in MAE between XGB-43 and XGB-37, paired by seed, sour lime (limón sutil); (a) all months, (b) shock months (monthly P75, MoM), (c) normal months. XGB-43 uses the 37 predictors of XGB-37 plus six lagged NLP features (avg_sentiment and n_noticias at lags 1, 3 and 6) with the same hyperparameters. Dots: individual seed pairs per test window (2021–2025) and pooled 2021–2025; diamond: mean over 10 seeds for the pooled 2021–2025 set with paired bootstrap 95% CI (months resampled within windows, B = 10,000). No mean or CI is shown per window. n/a = no shock month. XGB-43 was added post hoc: it is not part of the preregistered ranking and is not included in the Diebold–Mariano tests or the Holm correction. *Sweet lime:* EXPLORATORY (post-hoc) comparison: difference in MAE between XGB-43 and XGB-37, paired by seed, sweet lime (limón dulce); (a) all months, (b) shock months (monthly P75, MoM), (c) normal months. XGB-43 uses the 37 predictors of XGB-37 plus six lagged NLP features (avg_sentiment and n_noticias at lags 1, 3 and 6) with the same hyperparameters. Dots: individual seed pairs per test window (2021–2025) and pooled 2021–2025; diamond: mean over 10 seeds for the pooled 2021–2025 set with paired bootstrap 95% CI (months resampled within windows, B = 10,000). No mean or CI is shown per window. n/a = no shock month. XGB-43 was added post hoc: it is not part of the preregistered ranking and is not included in the Diebold–Mariano tests or the Holm correction.

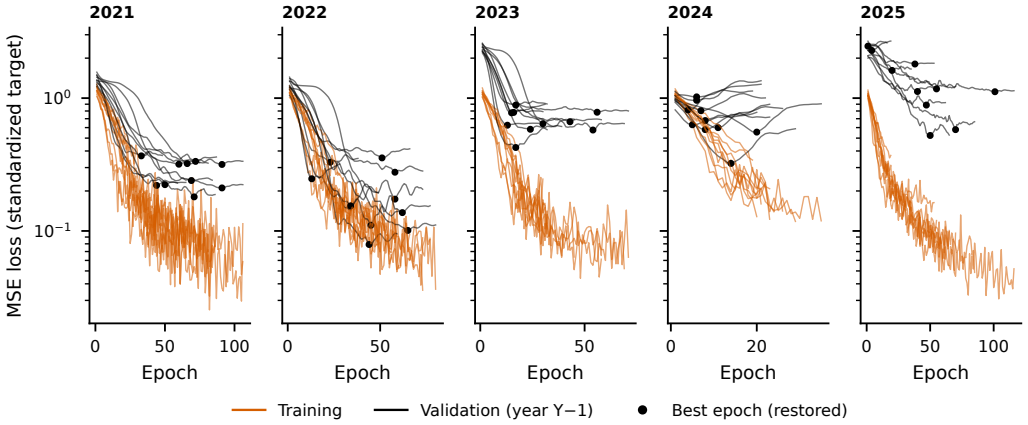


Figure 44. Sour lime

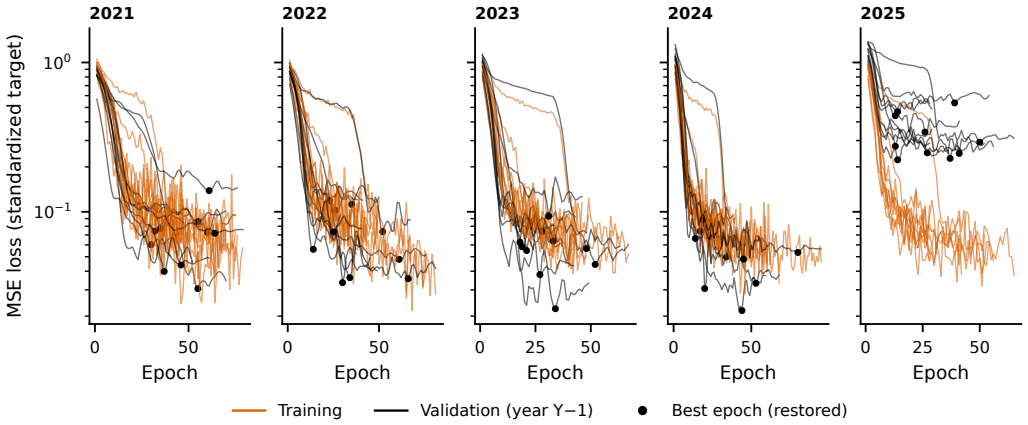


Figure 45. Sweet lime

Figure 46. *Sour lime:* Training and validation loss curves of GC3 for each test window and the 10 seeds (one line per seed), sour lime (limón sutil). Dots: epoch with minimum validation loss, whose weights are restored for forecasting. *Sweet lime:* Training and validation loss curves of GC3 for each test window and the 10 seeds (one line per seed), sweet lime (limón dulce). Dots: epoch with minimum validation loss, whose weights are restored for forecasting.

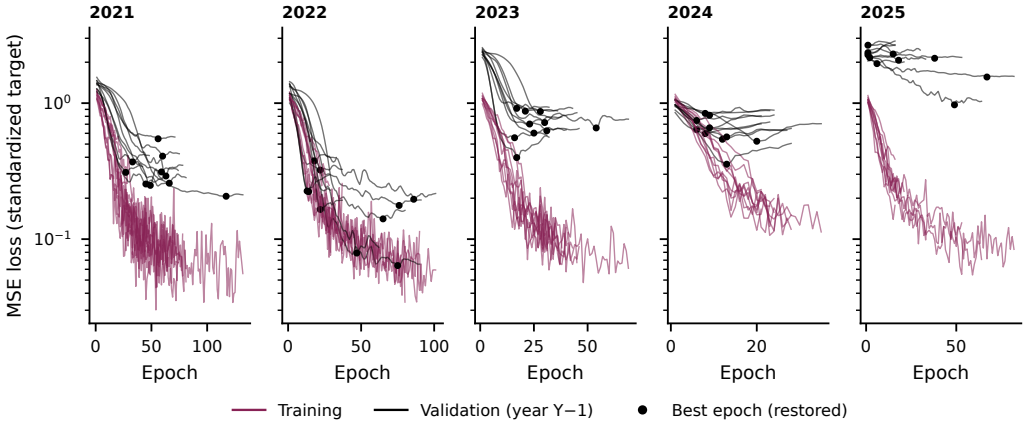


Figure 47. Sour lime

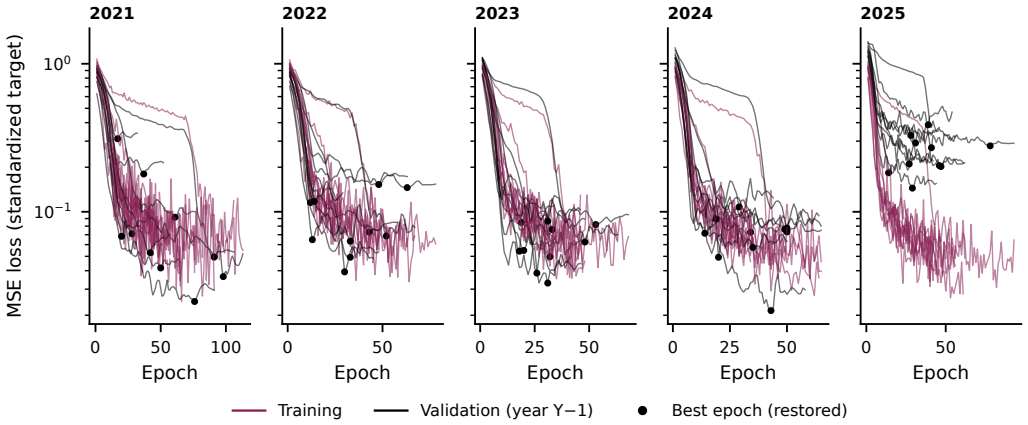


Figure 48. Sweet lime

Figure 49. *Sour lime:* Training and validation loss curves of GE for each test window and the 10 seeds (one line per seed), sour lime (limón sutil). Dots: epoch with minimum validation loss, whose weights are restored for forecasting. *Sweet lime:* Training and validation loss curves of GE for each test window and the 10 seeds (one line per seed), sweet lime (limón dulce). Dots: epoch with minimum validation loss, whose weights are restored for forecasting.

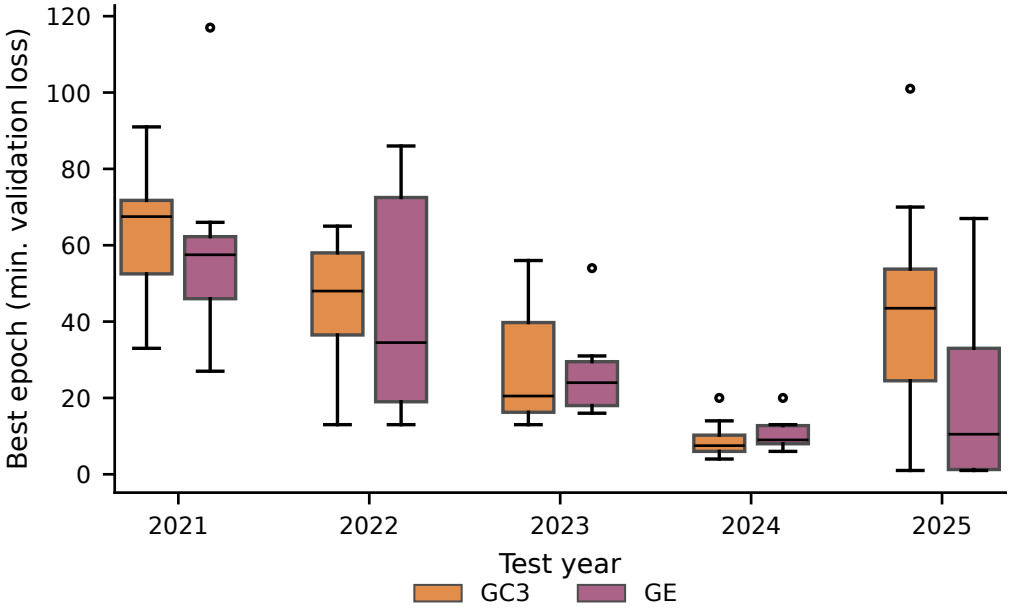


Figure 50. Sour lime

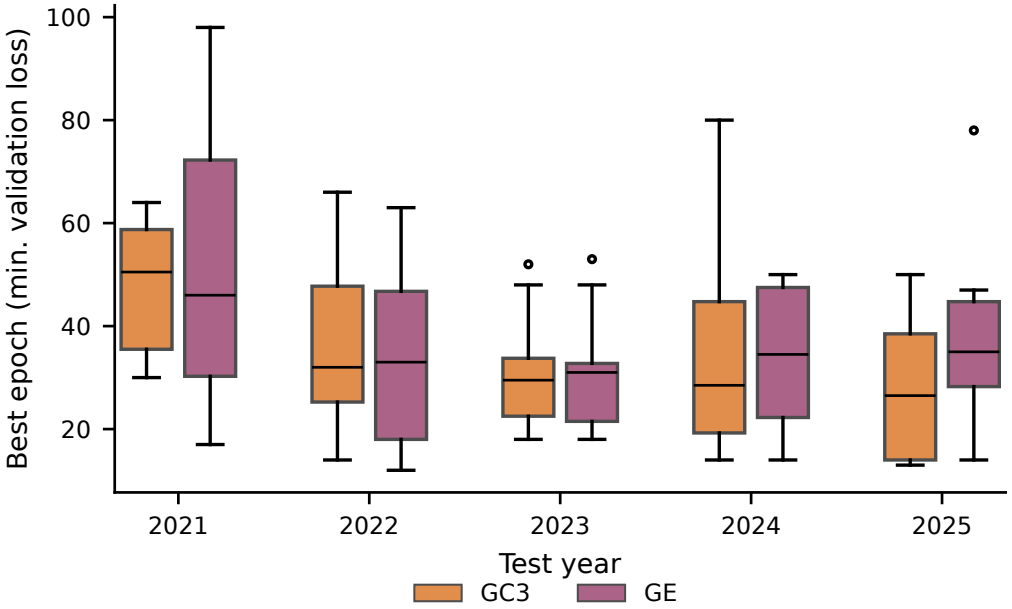


Figure 51. Sweet lime

Figure 52. *Sour lime:* Distribution over 10 seeds of the best (restored) epoch of GC3 and GE per test window, sour lime (limón sutil); early stopping on validation year Y−1, patience 15, maximum 300 epochs. *Sweet lime:* Distribution over 10 seeds of the best (restored) epoch of GC3 and GE per test window, sweet lime (limón dulce); early stopping on validation year Y−1, patience 15, maximum 300 epochs.

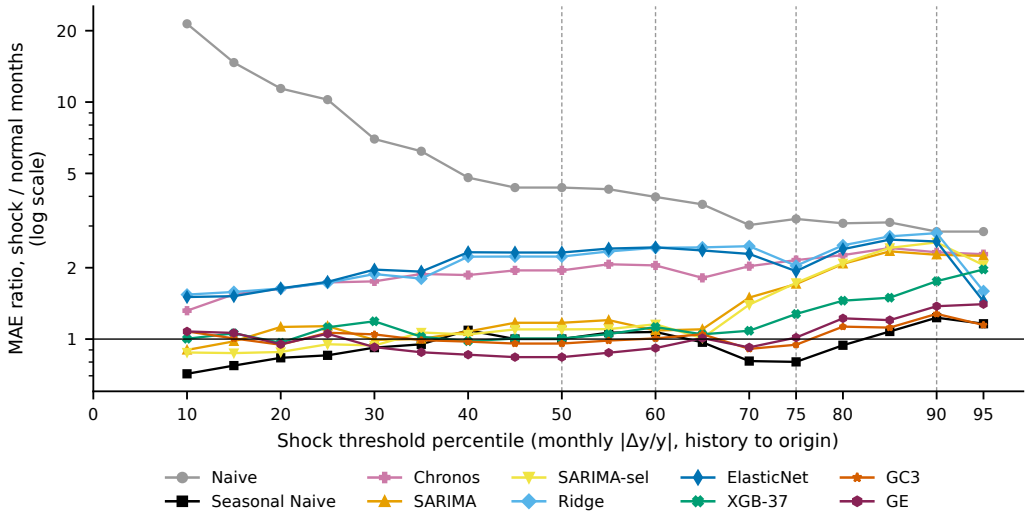


Figure 53. Sour lime

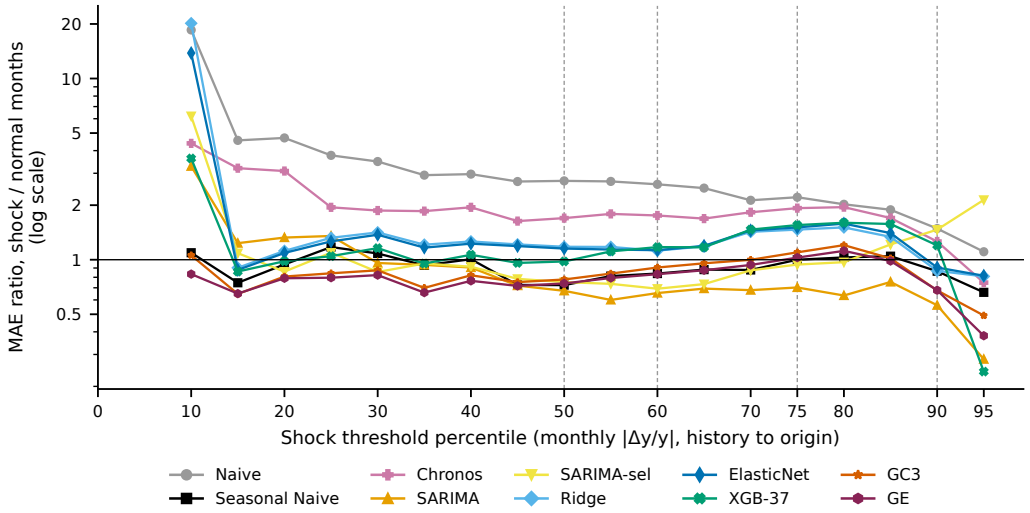


Figure 54. Sweet lime

Figure 55. *Sour lime:* Ratio of shock-month to normal-month MAE versus threshold percentile, sour lime (limón sutil), 2021–2025. Undefined at P0. Horizontal line: ratio = 1. Logarithmic y-axis, so ratios r and 1/r are equally far from 1. *Sweet lime:* Ratio of shock-month to normal-month MAE versus threshold percentile, sweet lime (limón dulce), 2021–2025. Undefined at P0. Horizontal line: ratio = 1. Logarithmic y-axis, so ratios r and 1/r are equally far from 1.

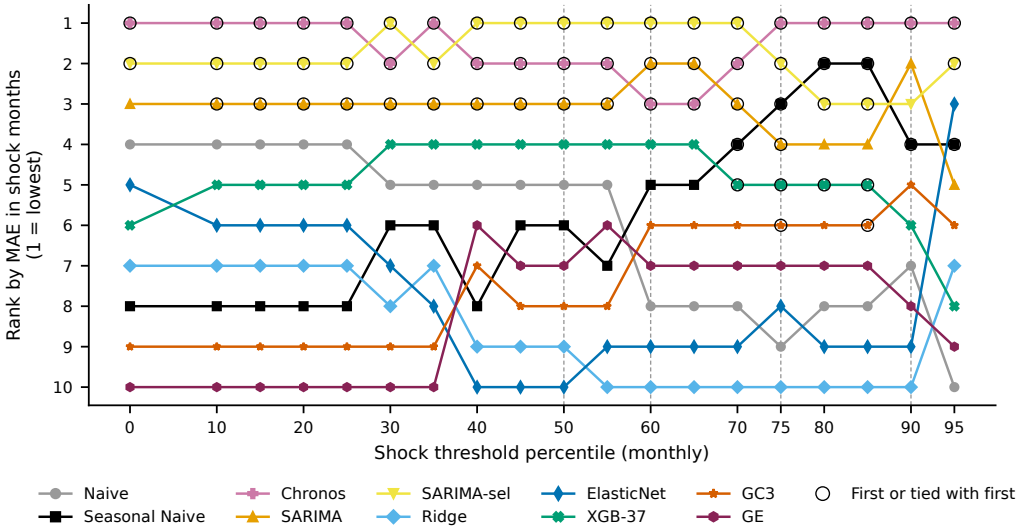


Figure 56. Sour lime

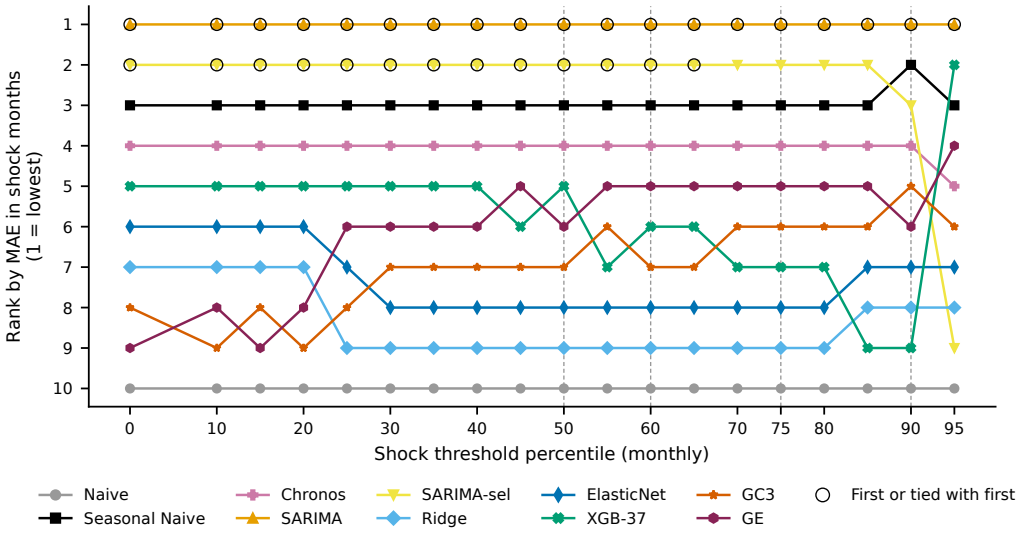


Figure 57. Sweet lime

Figure 58. *Sour lime:* Rank trajectories by MAE_shock across monthly threshold percentiles, sour lime (limón sutil), 2021–2025. Open circles: first place or tied with it (paired bootstrap). Dashed lines: P50, P60, P75, P90. *Sweet lime:* Rank trajectories by MAE_shock across monthly threshold percentiles, sweet lime (limón dulce), 2021–2025. Open circles: first place or tied with it (paired bootstrap). Dashed lines: P50, P60, P75, P90.

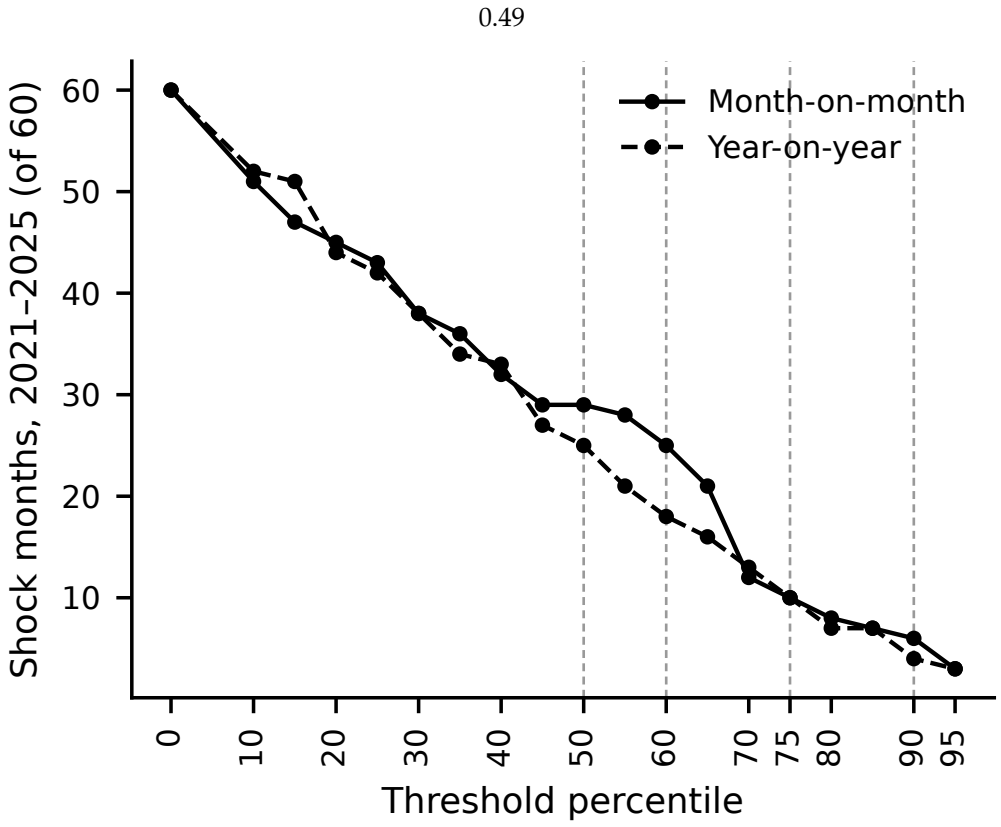


Figure 59. Sour lime

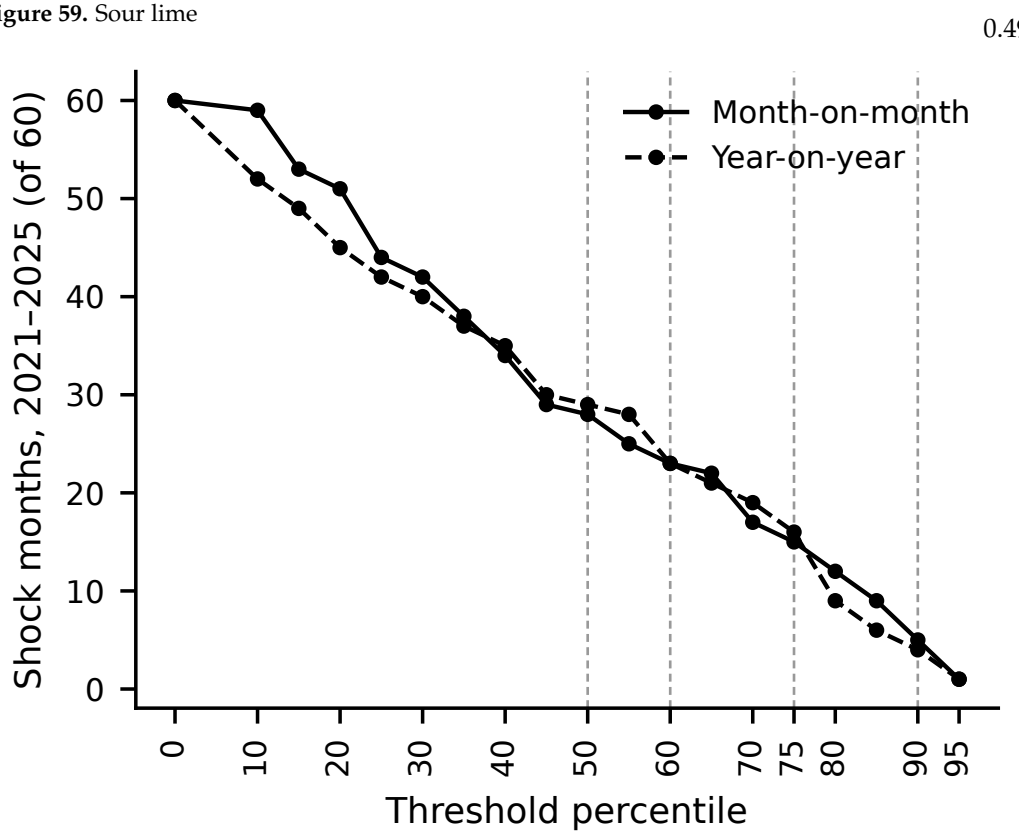


Figure 60. Sweet lime

Figure 61. *Sour lime:* Number of shock months in the pooled 2021–2025 test set versus threshold percentile, sour lime (limón sutil); solid: month-on-month change, dashed: year-on-year change. *Sweet lime:* Number of shock months in the pooled 2021–2025 test set versus threshold percentile, sweet lime (limón dulce); solid: month-on-month change, dashed: year-on-year change.

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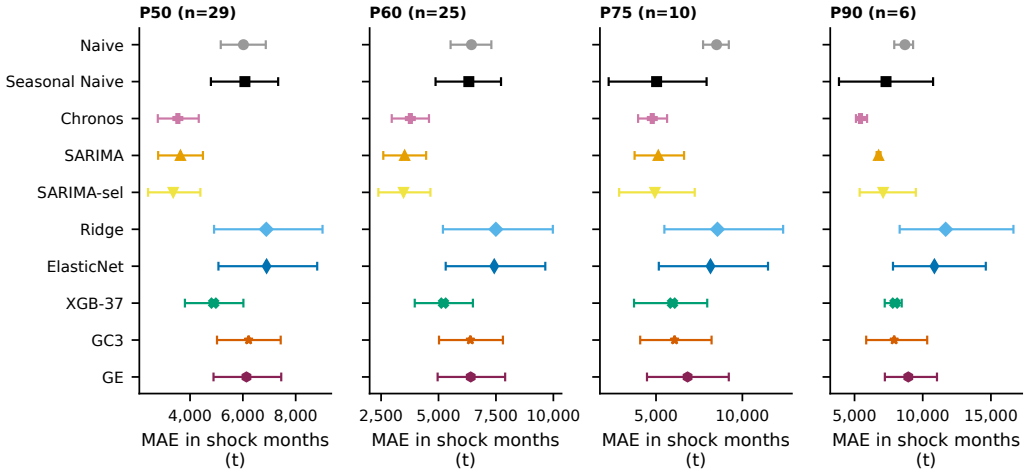


Figure 62. Sour lime

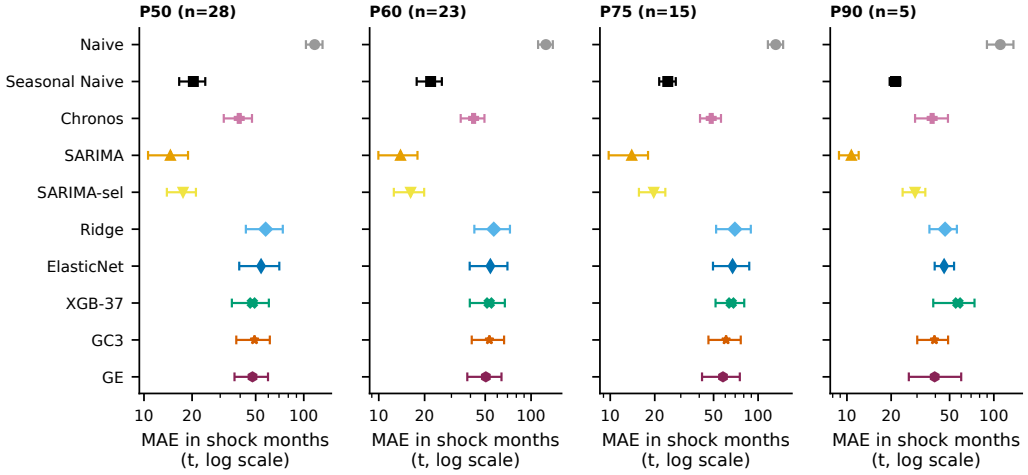


Figure 63. Sweet lime

Figure 64. *Sour lime:* MAE in shock months with paired bootstrap 95% confidence intervals (i.i.d. over months, stratified by window, B = 10,000) at the reporting thresholds P50, P60, P75 and P90 (monthly), sour lime (limón sutil), 2021–2025. *Sweet lime:* MAE in shock months with paired bootstrap 95% confidence intervals (i.i.d. over months, stratified by window, B = 10,000) at the reporting thresholds P50, P60, P75 and P90 (monthly), sweet lime (limón dulce), 2021–2025. Logarithmic x-axis.

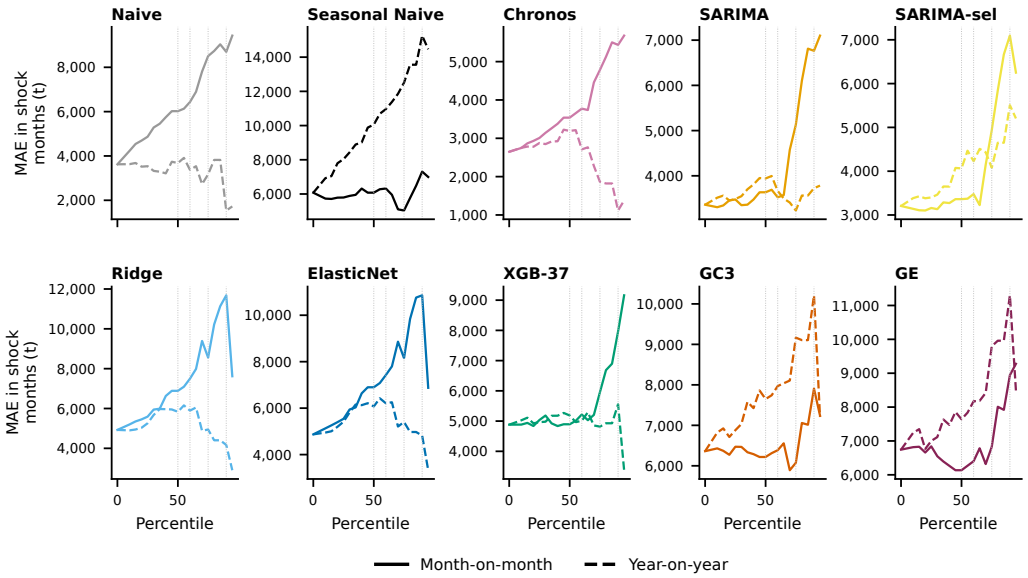


Figure 65. Sour lime

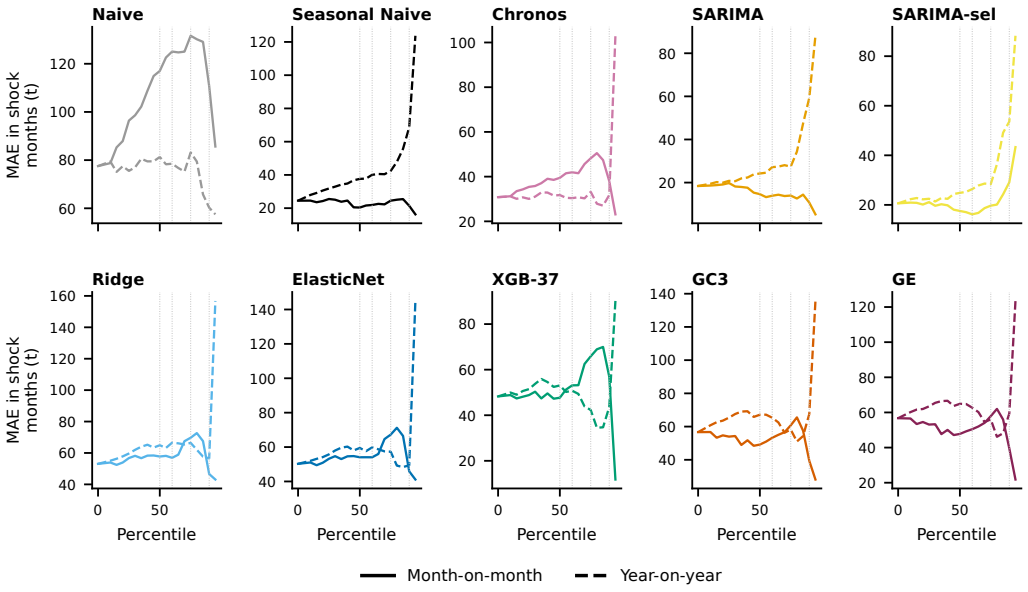


Figure 66. Sweet lime

Figure 67. *Sour lime:* MAE in shock months versus threshold percentile for each model, month-on-month (solid) versus year-on-year (dashed) shock definition, sour lime (limón sutil), 2021–2025. Each panel has its own y-axis. *Sweet lime:* MAE in shock months versus threshold percentile for each model, month-on-month (solid) versus year-on-year (dashed) shock definition, sweet lime (limón dulce), 2021–2025. Each panel has its own y-axis.

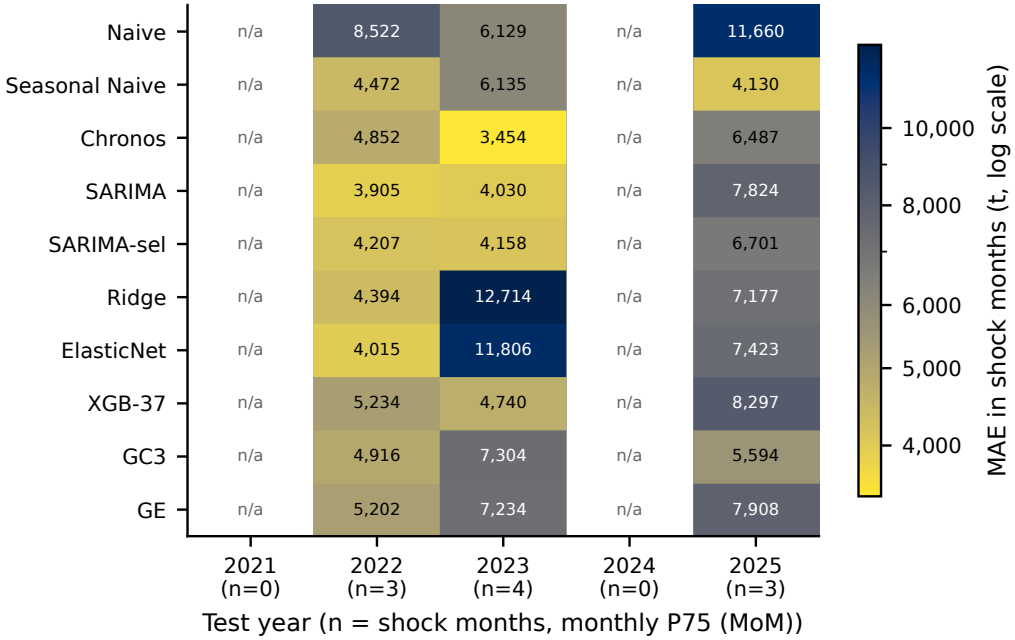


Figure 68. Sour lime

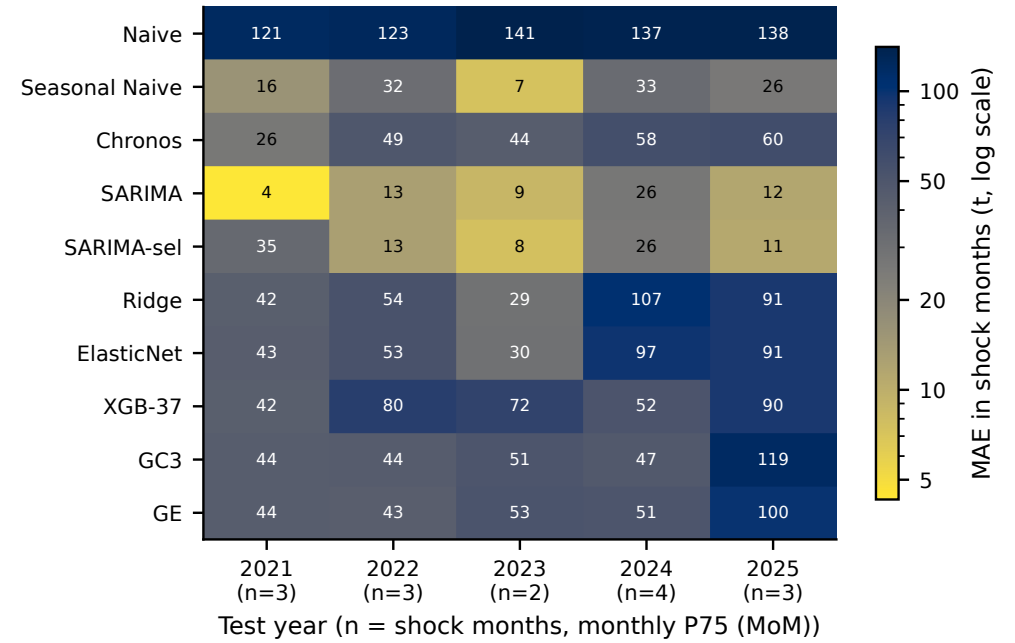


Figure 69. Sweet lime

Figure 70. *Sour lime:* MAE in shock months (monthly P75) by model and test window, sour lime (limón sutil). n/a = no shock month in that window. *Sweet lime:* MAE in shock months (monthly P75) by model and test window, sweet lime (limón dulce). n/a = no shock month in that window.

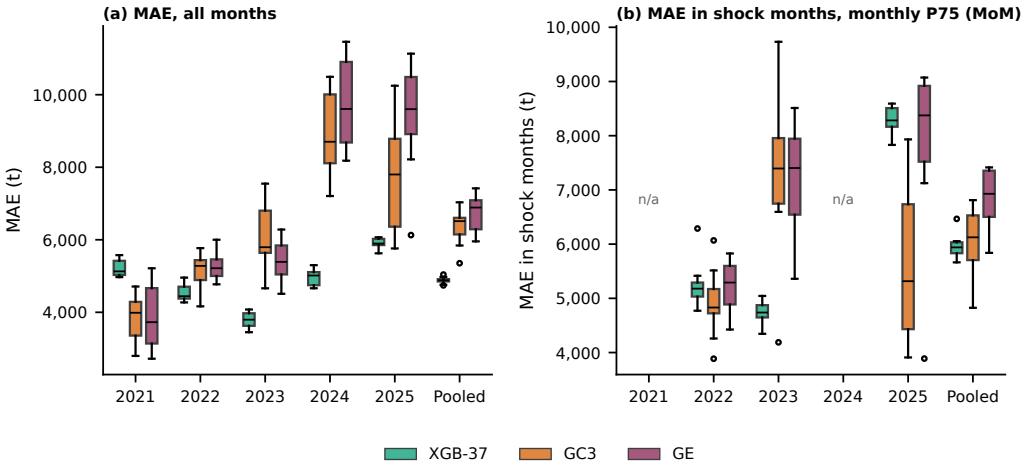


Figure 71. Sour lime

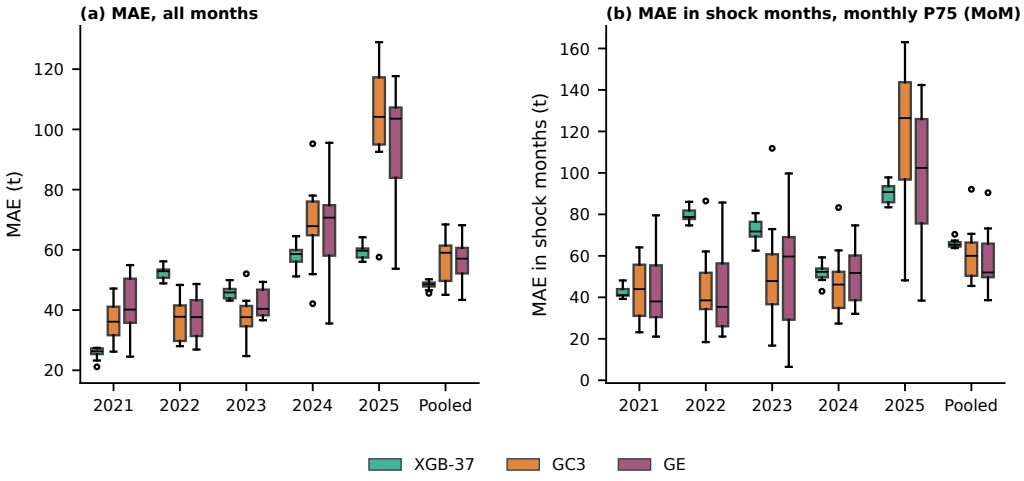


Figure 72. Sweet lime

Figure 73. *Sour lime:* Distribution over the 10 training seeds of (a) MAE and (b) shock-month MAE (monthly P75) per test window and pooled 2021–2025, for the seeded models, sour lime (limón sutil). *Sweet lime:* Distribution over the 10 training seeds of (a) MAE and (b) shock-month MAE (monthly P75) per test window and pooled 2021–2025, for the seeded models, sweet lime (limón dulce).

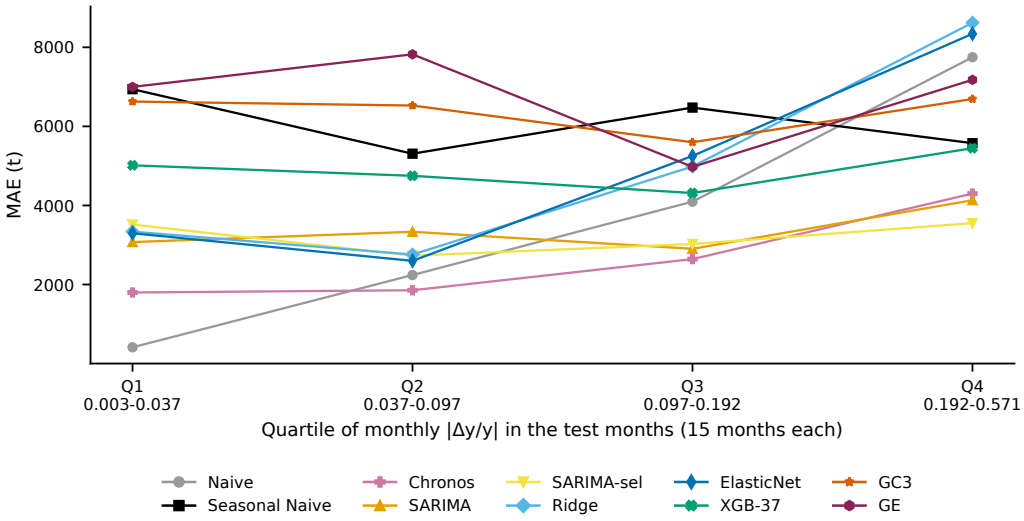


Figure 74. Sour lime

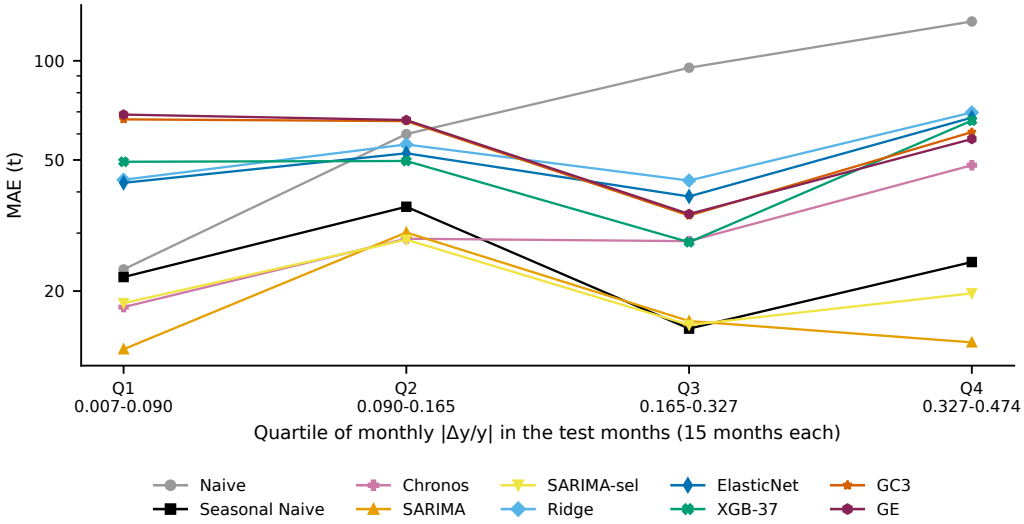


Figure 75. Sweet lime

Figure 76. *Sour lime:* MAE by quartile of the absolute month-on-month relative change $|\Delta y/y|$ of the test months, sour lime (limón sutil), pooled 2021–2025. *Sweet lime:* MAE by quartile of the absolute month-on-month relative change $|\Delta y/y|$ of the test months, sweet lime (limón dulce), pooled 2021–2025.

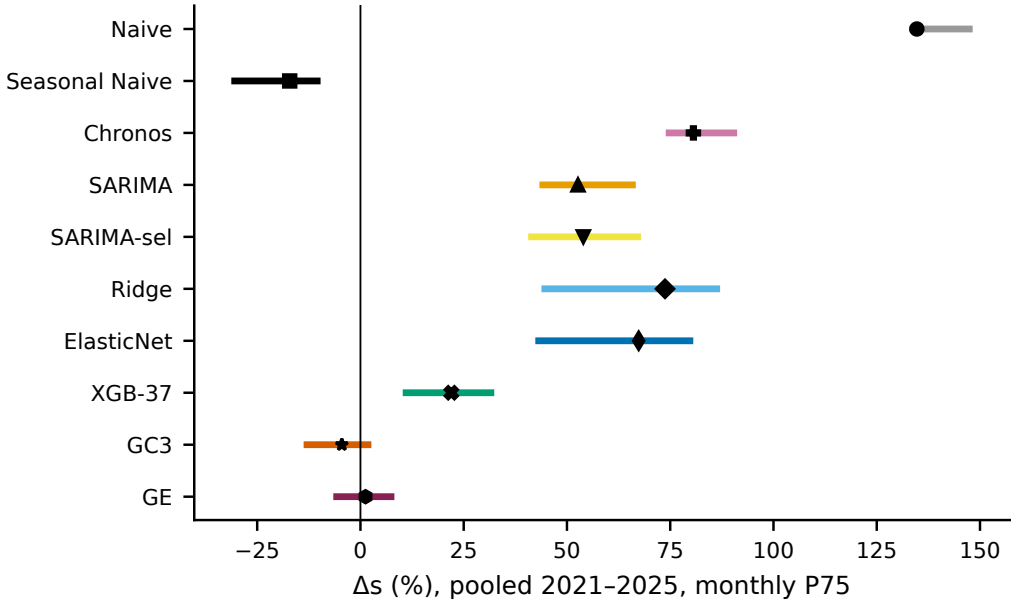


Figure 77. Sour lime

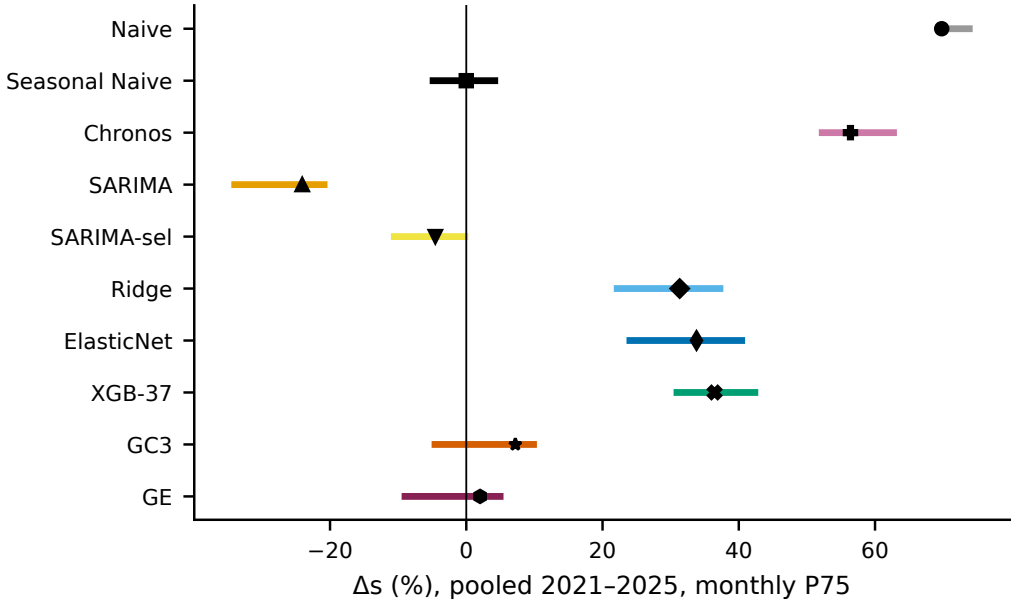


Figure 78. Sweet lime

Figure 79. *Sour lime:* Δs with all shock months (marker) and its range when one shock month is removed at a time (bar), sour lime (limón sutil), pooled 2021–2025, monthly P75. *Sweet lime:* Δs with all shock months (marker) and its range when one shock month is removed at a time (bar), sweet lime (limón dulce), pooled 2021–2025, monthly P75.

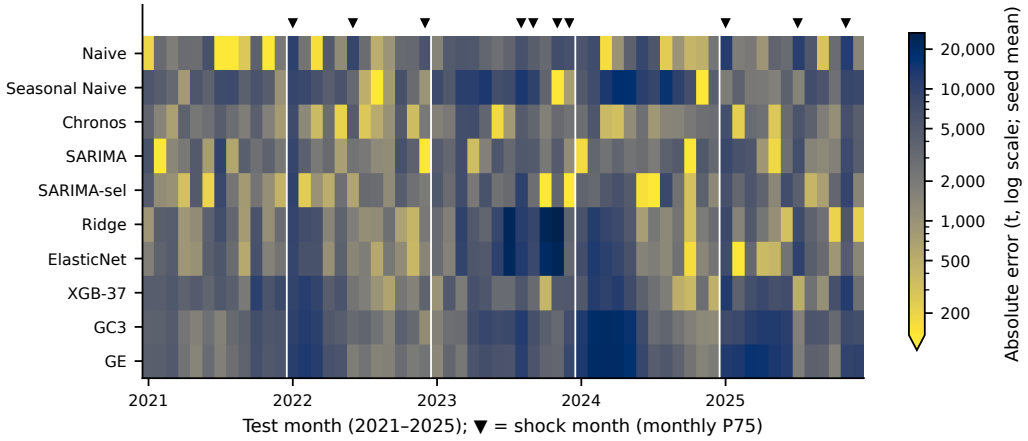


Figure 80. Sour lime

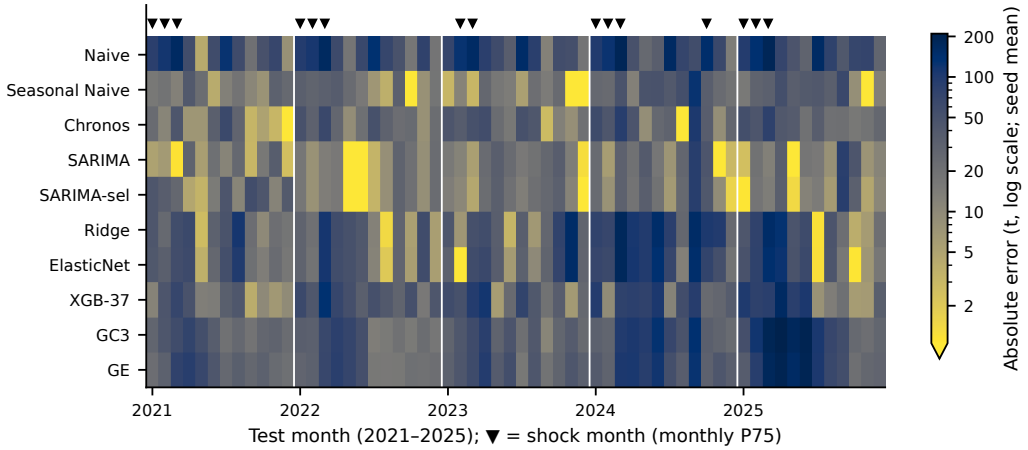


Figure 81. Sweet lime

Figure 82. *Sour lime:* Month-by-month absolute forecast error of each model (mean over seeds for seeded models), sour lime (limón sutil), test months 2021–2025. *Sweet lime:* Month-by-month absolute forecast error of each model (mean over seeds for seeded models), sweet lime (limón dulce), test months 2021–2025.

Naive and the pretrained model. This agrees with evidence that statistical and regularised methods are demanding benchmarks in short series [1,3,5] and indicates that a specialised architecture should always be compared with them, and with pretrained models, before being proposed for operational use.

Overall versus shock-conditional evaluation. Restricting the evaluation to shock months did not favour the DualLSTM. Its low Δ_s in sour lime (-4.5% for GC3) reflects a high error in all months, not a low error in shock months: Δ_s is a ratio, and a model that is poor in every month can show a small penalty. The same applies to the Seasonal Naive ($\Delta_s = -17.1\%$). Reporting Δ_s alone would have ranked these models as the most resilient, whereas in absolute terms Chronos, SARIMA and SARIMA-sel were better in shock months as well. We therefore recommend reporting Δ_s only together with the absolute errors in shock and normal months. The shock-only comparisons rest on 10 and 15 months and have low power, so the ranking in shock months should be read as a complementary diagnostic and not as evidence of intrinsic resilience.

Contribution of news text. News features did not improve the DualLSTM: they increased the error in sour lime and made no difference in sweet lime. The only favourable signal, a lower shock error of XGB-43 in sweet lime, comes from a post hoc analysis outside the ranking. Overall, there is not enough evidence to claim that the text channel improves accuracy in this setting. Sentiment was computed zero-shot with a general-purpose model, and a model fitted to agricultural news could behave differently.

Limitations. The evaluation uses two related national series and five retrospective test windows (60 pooled months), which are not independent test sets; the test year 2025 had already been inspected in earlier experiments, and the order of the fixed sour-lime SARIMA was adopted after seeing it (Section 2). Shock months are few (10 and 15), the confidence intervals are wide and several pairwise differences cannot be separated from zero. The networks were run with fixed hyperparameters and no search, which may disadvantage them relative to models whose penalties were re-chosen in each window. Chronos uses only the univariate history, whereas the DualLSTM receives multimodal variables, so the comparison concerns predictive performance and not equal information sets. The attention weights of the DualLSTM should not be interpreted as causal relations between climate, emergencies or news and production [46]. [PENDIENTE: rewrite this section once the final tables and the Holm results are fixed.]

5. Conclusions

A DualLSTM with attention that separates the production history from the lagged exogenous context did not consistently outperform statistical, regularised, tree-based or pretrained benchmarks in the monthly forecasting of sour lime and sweet lime production in Peru under a rolling-origin evaluation with 60 test months per crop. The best model depended on the crop: Chronos, SARIMA and SARIMA-sel in sour lime, and SARIMA in sweet lime, which was the only model that was first or tied at all four shock thresholds. The low shock penalty of the DualLSTM was explained by high errors in all months, so Δ_s must be reported together with the absolute errors. News-derived variables showed no consistent benefit. Future work should extend the analysis to more crops and regions, longer evaluation periods and independently documented external events, and should test networks with hyperparameter search.

Author Contributions: Conceptualization, F.Y.A.S.S., A.C.J. and E.T.-C.; methodology, F.Y.A.S.S. and E.T.-C.; software, F.Y.A.S.S.; validation, F.Y.A.S.S. and A.C.J.; formal analysis, F.Y.A.S.S.; investigation, F.Y.A.S.S. and A.C.J.; data curation, F.Y.A.S.S.; writing—original draft preparation, F.Y.A.S.S.; writing—review and editing, A.C.J. and E.T.-C.; visualization, F.Y.A.S.S.; supervision, E.T.-C. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The national production series were built from the MIDAGRI SISAGRI Regional Productive Profile dashboard [29], climate data from NASA POWER, emergency records from INDECI and news from Agraria.pe. The frozen tables used in this study, with their SHA-256 hashes, and the code are available at https://github.com/YeralD21/machine-learning-multimodal--agro-nlp-clima- [PENDIENTE: add the release tag after pushing the final code].

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Prácticas metodológicas de los artículos semilla

Las Tablas A1 y A2 resumen la práctica que este estudio adopta de cada artículo semilla y la debilidad que evita. La síntesis procede de la lectura del texto completo de 18 artículos.

Table A1. Prácticas adoptadas y debilidades evitadas (parte 1).

Artículo	Práctica adoptada	Debilidad evitada
Gu et al. [7]	Doble entrada para serie y clima; experimento aislado del aporte de las variables climáticas	Cuatro observaciones de prueba, usadas además para elegir la ventana
Yin et al. [8]	Ablación limpia de componentes del modelo	Ventana elegida sobre seis puntos de prueba
Bao et al. [9]	Ablación por conjuntos de variables; análisis en fechas de *shock*	Variables anuales interpoladas a frecuencia diaria
Praveenkumar et al. [47]	Pruebas de estacionariedad y no linealidad antes de modelar	Una sola corrida, sin línea base ingenua ni pruebas de significancia
Zhao et al. [16]	Error separado por periodos de alta y baja volatilidad; sensibilidad a la partición	Cifras inconsistentes entre tablas y resumen
Ray et al. [13]	Batería de diagnóstico, Diebold-Mariano y MASE frente al predictor ingenuo	Doce puntos de prueba
Avinash et al. [14]	Diez repeticiones, varios horizontes y Diebold-Mariano por pares	Sin ablación ni líneas base ingenuas
Manogna et al. [10]	Diebold-Mariano sobre varias funciones de pérdida	Conclusiones del texto que no coinciden con las tablas
Nayak et al. [15]	Clima de NASA POWER; ablación por variable exógena; MASE	Sin conjunto de validación ni pruebas de significancia

Table A2. Prácticas adoptadas y debilidades evitadas (parte 2).

Artículo	Práctica adoptada	Debilidad evitada
An et al. [48]	Ablación del índice de sentimiento con Diebold-Mariano frente a cada comparador	Metamodelo ajustado con predicciones sobre el conjunto de prueba
Ryu et al. [49]	Diseño con y sin datos no estructurados; reporte de un resultado no significativo	Configuraciones seleccionadas sobre el año de prueba
Sediyono et al. [12]	Umbral de anomalía derivado de la distribución del error	Partición descrita de forma inconsistente; anomalías sin etiquetas reales
Suliman et al. [**?**]	Validación de origen móvil; error por periodo pico y normal; simulación de inventario	Datos sintéticos; tamaños de partición sin reportar
Paudel et al. [50]	Diez modelos por semilla; explicaciones contrastadas con expertos; código abierto	Evaluación de interpretabilidad con cuatro expertos
Fırat y Sarıkaya [51]	Escalado ajustado solo con entrenamiento; ablación; intervalos por *bootstrap*	Variable derivada del propio precio objetivo
Sabo et al. [52]	Líneas base simples; validación por año no visto; reporte de resultado negativo	Sin pruebas de significancia ni tablas numéricas
Chergui [53]	Mismos conjuntos de prueba en todos los experimentos; descripción del área con mapa	Muestras de prueba de cuatro a siete años, sin intervalos
Jauhar et al. [54]	Concordancia entre tres métodos de explicabilidad	Serie pronosticada generada de forma sintética

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